



LAB MANUAL

HCI & COMPUTER GRAPHICS

CSC-309

Fall-2025

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Section: 5C

DEPARTMENT OF SOFTWARE ENGINEERING



LAB MANUAL

HCI & COMPUTER GRAPHICS

This is to certify that Mr/Ms. Aasim Nazim S/o/D/o Muhammad Nazim Khan have ID no. BSE-23F-146 has successfully completed Laboratory work during Fall Semester 2025.

LAB INSTRUCTOR

MS. LARAIB WAHEED

Signature

S. No	Objectives	Remarks
01	Introduction to Computer Graphics & HCI	
02	Grid-Based Layouts in UI Designs	
03	Typography & Color Theory in UI Designs	
04	UI Elements (Forms & Buttons)	
05	Sider Menu Bar & Scrolling Behavior	
06	Interaction between screens by using prototyping in Figma.	
07	Introduction to Blender & Installation	
08	Blender Basics (Object Mode & Mesh)	
09	Blender (Edit Mode) Basic Tools	
10	Advanced Mesh Editing by Edit Mode	

LAB 01

Introduction to HCI & Computer Graphics

Objective:

- Introduce **HCI & Computer Graphics** concepts.
- Explain **Wireframes** and their types.
- Get students familiar with **Figma interface**.
- Teach them to create a **very simple wireframe** in Figma.

Introduction:

Computer Graphics:

Computer graphics is the branch of computer science that deals with creating, storing, and manipulating visual content (images, shapes, animations) using computers.

Applications

- UI/UX Design
- Games
- Movies / Animation
- Data Visualization

Wireframes:

A wireframe is a blueprint of a digital product's screen showing the layout of elements without actual design details (colors, fonts, etc.).

Why Wireframes:

- Saves time before detailed design.
- Focus on structure & user flow first.
- Easy to discuss and modify with team.

Types of Wireframes:

1. Low-Fidelity Wireframe

- Simple black & white layout.
- Basic shapes (rectangles, lines).
- No styling, no color.
(Perfect for early-stage idea validation)

2. Mid-Fidelity Wireframe

- Little more detailed than low-fi.
- Clear separation of sections, some text.
(Used for refining user flows)

3. High-Fidelity Wireframe

- Almost looks like final UI.
- Includes colors, images, actual content.
(Used before final development)

Task 1:

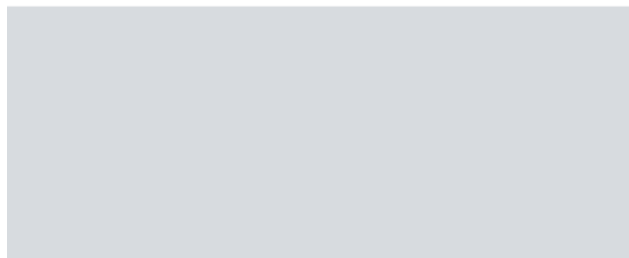
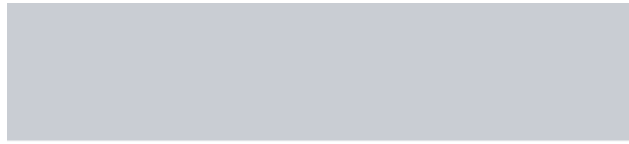
Create a simple basic wireframe which includes (Header, Footer, Title & Button shape) and also show basic alignment and grouping.

Task 2:

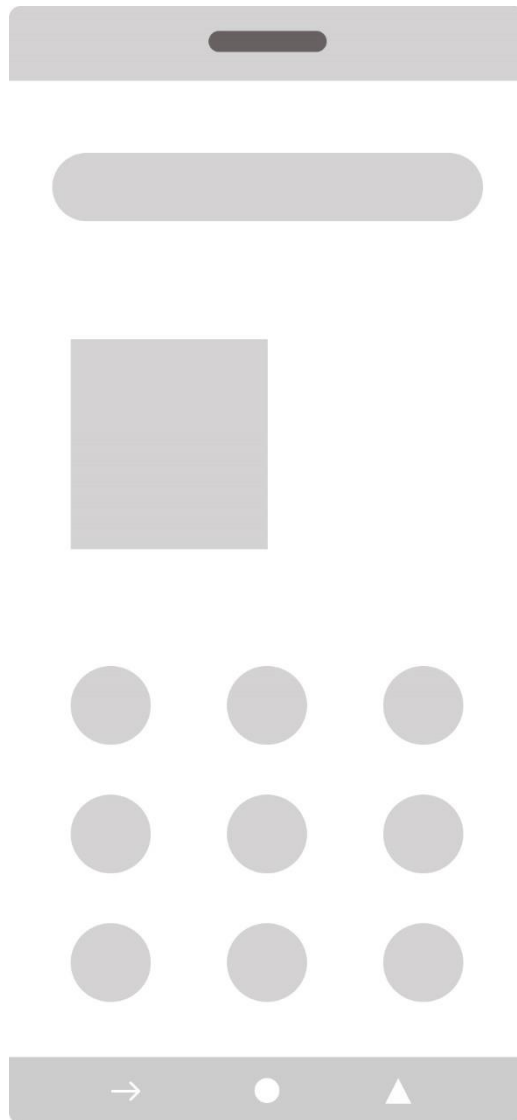
Create a **Wireframe for a Mobile App OR Home Screen:**

Paste screen shots of both tasks and Figma link (with anyone with link) access.

Task 1:



Task 2:



LAB 02

Grid-Based Layouts in UI Design

Objective:

- Understand the role of grids in creating structured and balanced user interfaces.
- Learn how to apply grid systems (rows/columns) in UI design tools like Figma.
- Design a simple **mobile app screen** using a 12-column grid system for alignment and consistency.

Introduction:

What is a Grid?

- A grid is an invisible structure made of rows and columns.
- It helps designers organize elements neatly, providing balance, alignment, and readability.

Why Grids in UI Design?

- Maintain consistency across screens.
- Create visual harmony (elements look aligned and balanced).
- Make designs more user-friendly and professional.

Types of Grids:

- Fixed Grid → Fixed width, works for specific screen sizes.
- Fluid Grid → Flexible, adjusts with screen size (responsive design).
- Modular Grid → Divides screen into modules (both rows & columns), good for dashboards.

12-Column Grid in Mobile UI

- The 12-column system is the industry standard (used in Bootstrap, Material Design).
- Provides flexibility → you can span elements across 2, 3, 4, or 6 columns.

- Ensures consistent spacing and alignment.

Example in a 12-Column Grid:

- **1 column** → a small button or icon.
- **2–3 columns** → a sidebar or small card.
- **6 columns** → a main content card or image.
- **12 columns** → a full-width banner or header.

Grid System Terminology

1. Column

- Vertical divisions of the grid (e.g., 12 columns).
- All design elements are placed inside these columns.

2. Row

- Horizontal divisions of the grid.
- Rows + columns together form the complete grid system.

3. Gutter

- The **space between columns**.
- It prevents elements from sticking too close to each other.
- Example: The gap between two text blocks or two images = gutter.

4. Margin

- The **outer empty space** on the left and right side of the layout.
- It ensures content does not touch the screen edges.

5. Module

- The small rectangular unit formed by the intersection of a row and a column.

6. Baseline Grid

- Horizontal lines used mainly for text alignment.
- Helps keep typography consistent.
- A **baseline grid** is a set of **equally spaced horizontal lines**.
- Mainly used for **text alignment**.
- Ensures that headings, paragraphs, and captions all sit neatly on the same invisible line.
- Keeps **typography consistent and readable** across the design.

Example: In Google Docs or MS Word, when you set **line spacing = 1.5**, text lines follow an invisible baseline grid.

- **Modules** = “boxes” for layout.
- **Baseline grid** = “lines” for text.

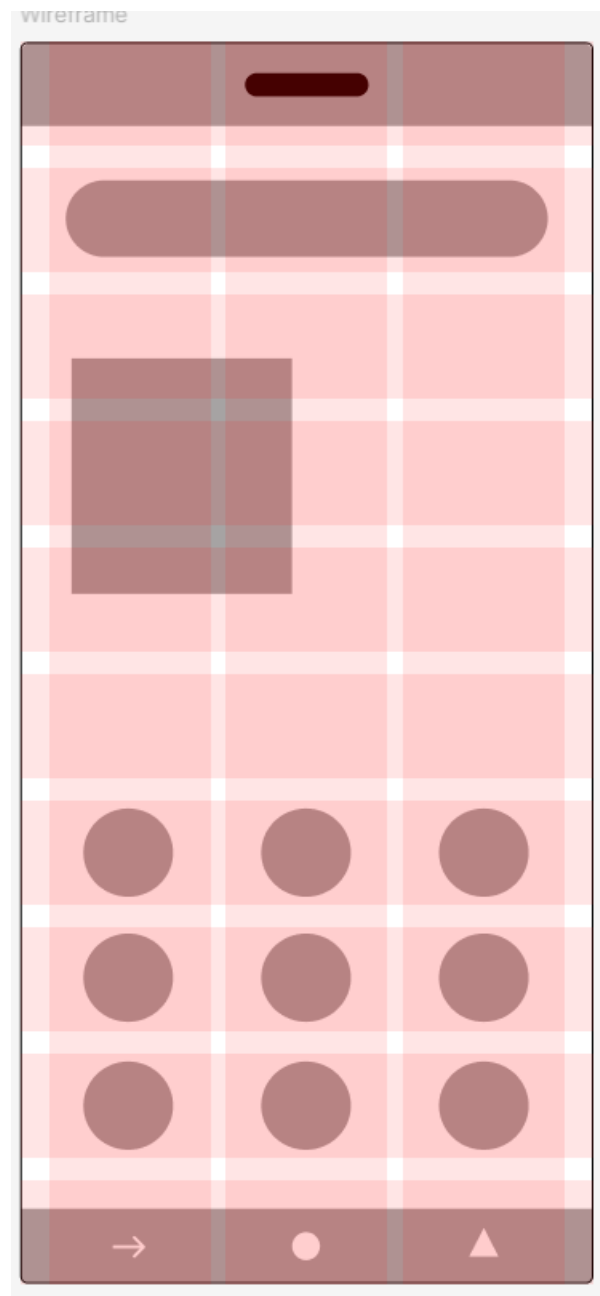
7. Spacing vs Padding

- **Spacing**: The distance *between* two different elements.
- **Padding**: The inner space *inside* an element (like space between button text and button border).

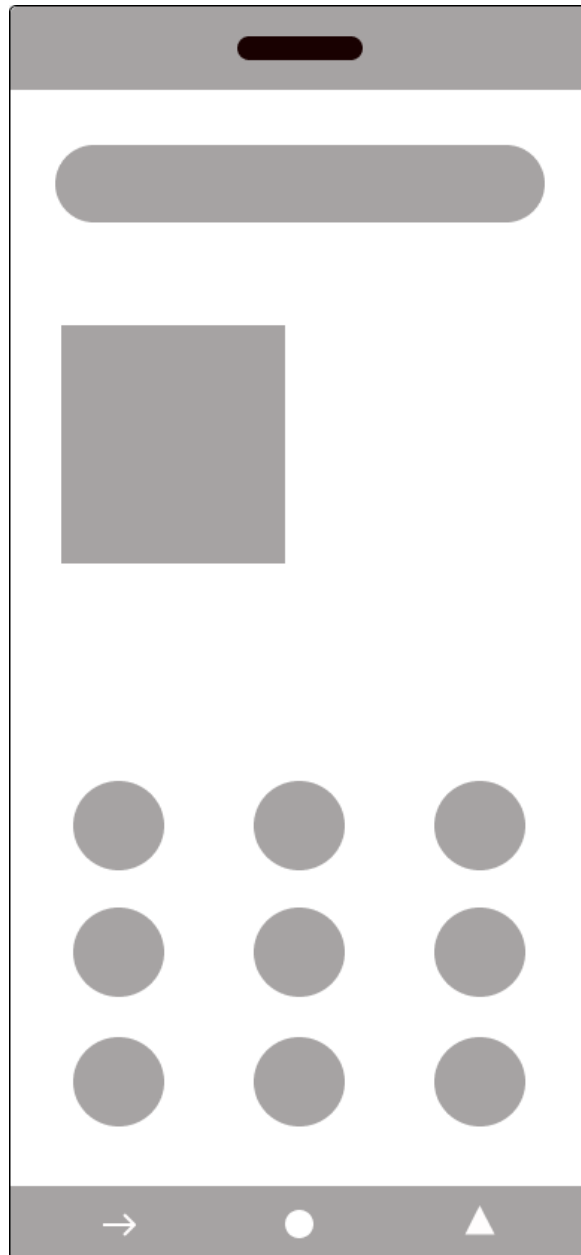
Task 1:

- Re-design the Home Screen Wireframe that you created earlier by using a Grid System.
- Apply margins and gutters properly.
- Export two jpeg versions of the screen:
 1. With grid visibility (show the grid lines in the screenshot).
 2. Without grid visibility (clean version of the wireframe).

Grid View:



Wireframe:

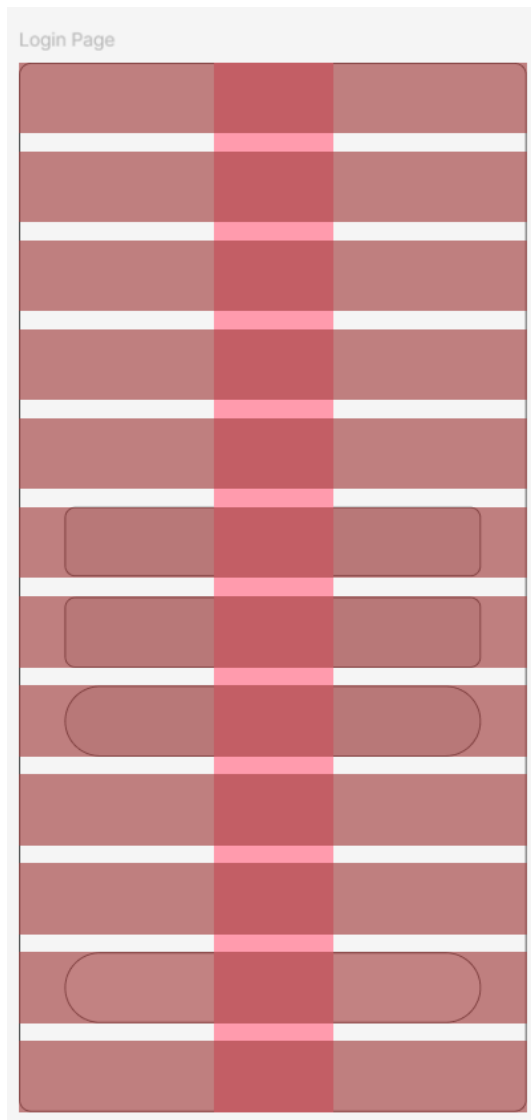


Task 2:

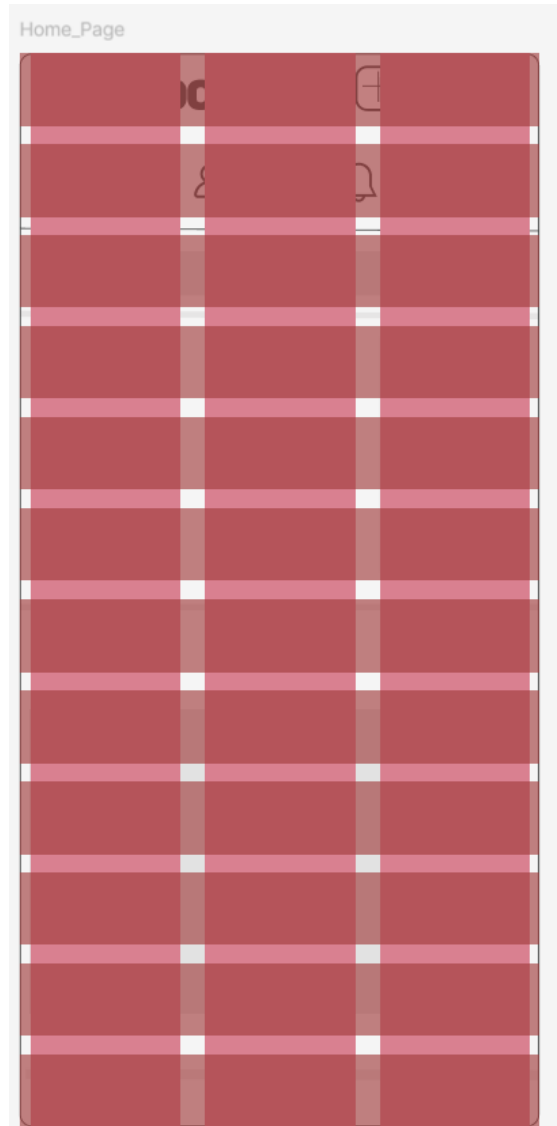
- Select any mobile application (e.g., Foodpanda, Daraz, Instagram, or any other).
- Create wireframes of minimum 3 screens from that app by using a Grid System.
- Apply margins and gutters properly.
- In a word file you need to submit:
 1. Screenshots of the actual app (for reference).
 2. Screen Shot of Wireframe with grid visible.
 3. Wireframe with grid invisible.

Paste screen shots of both tasks and Figma link (with anyone with link) access.

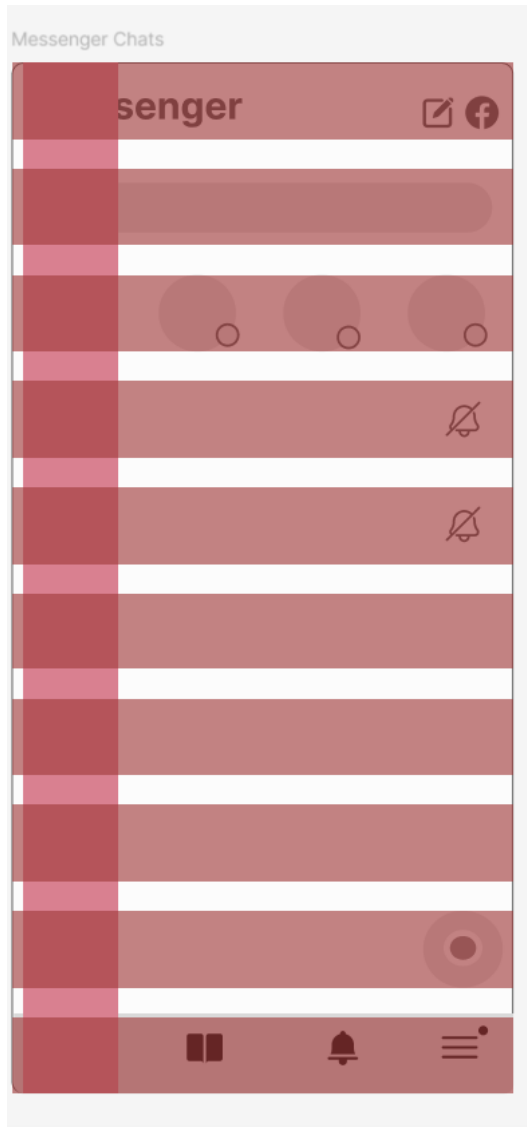
Grid View (Page-1)



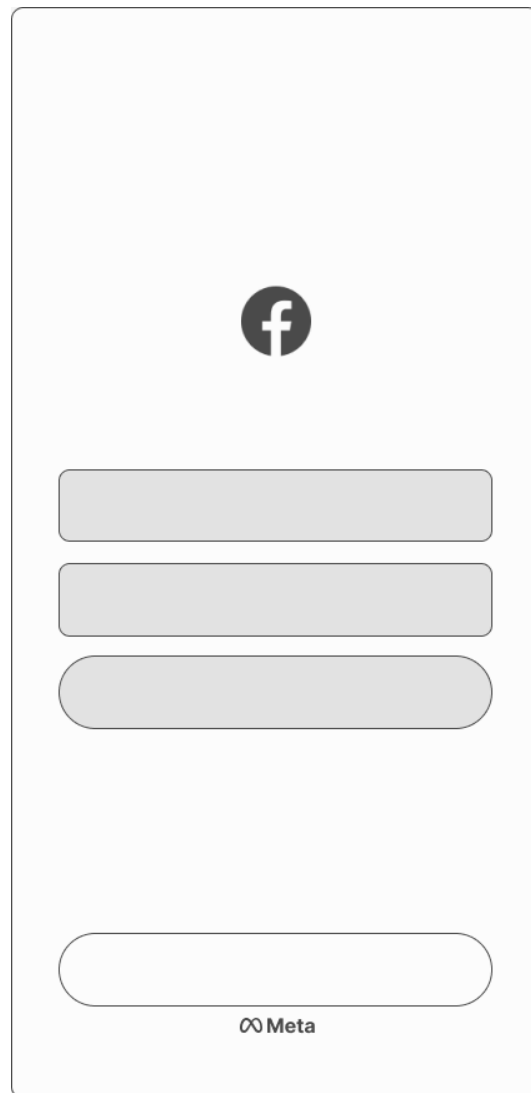
Grid View (Page-2)



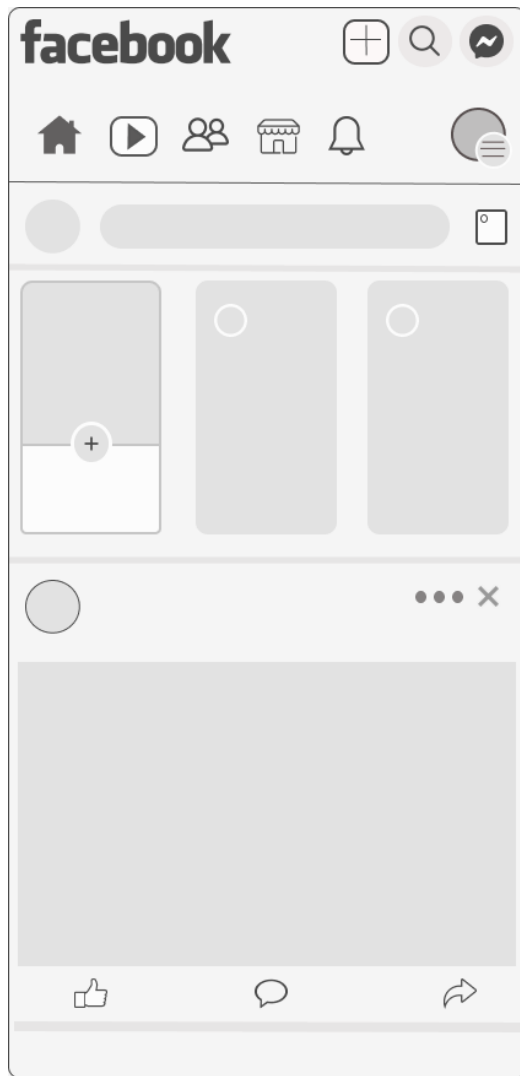
Grid View (Page-1)



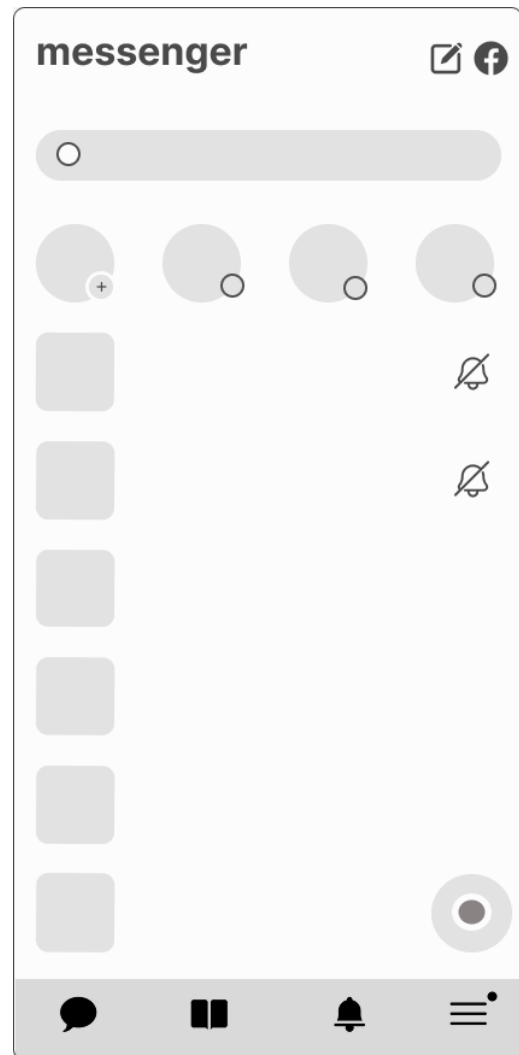
Wireframe: Page-1



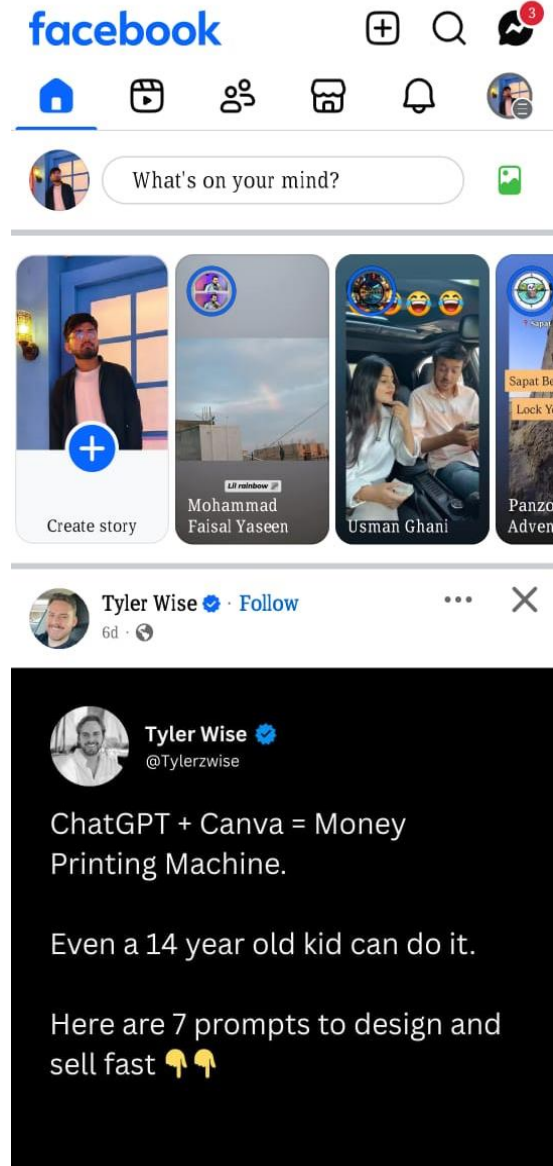
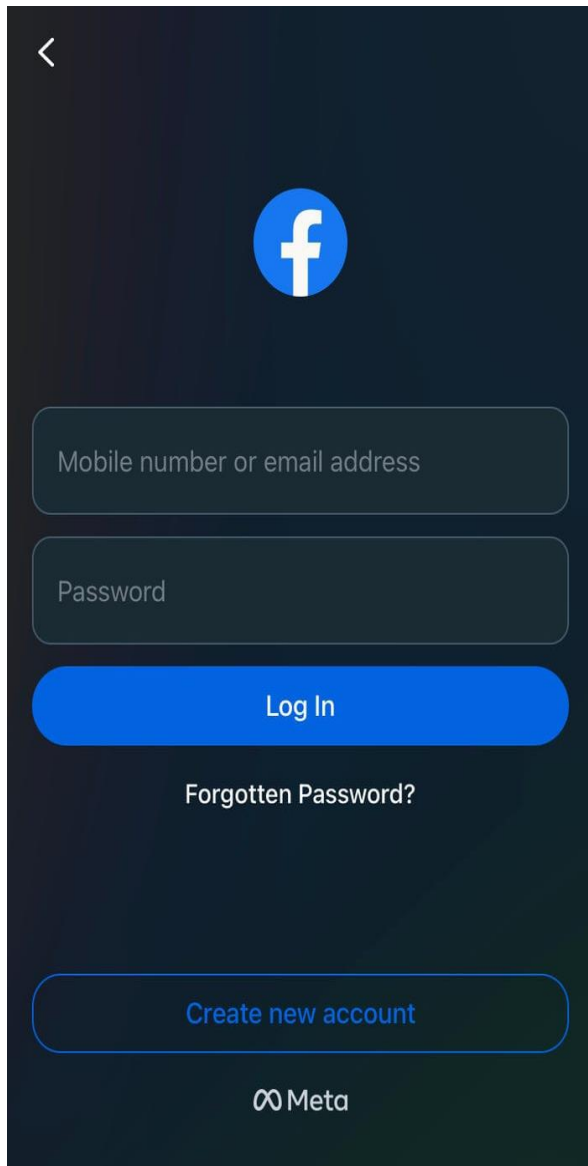
Wireframe: Page-2

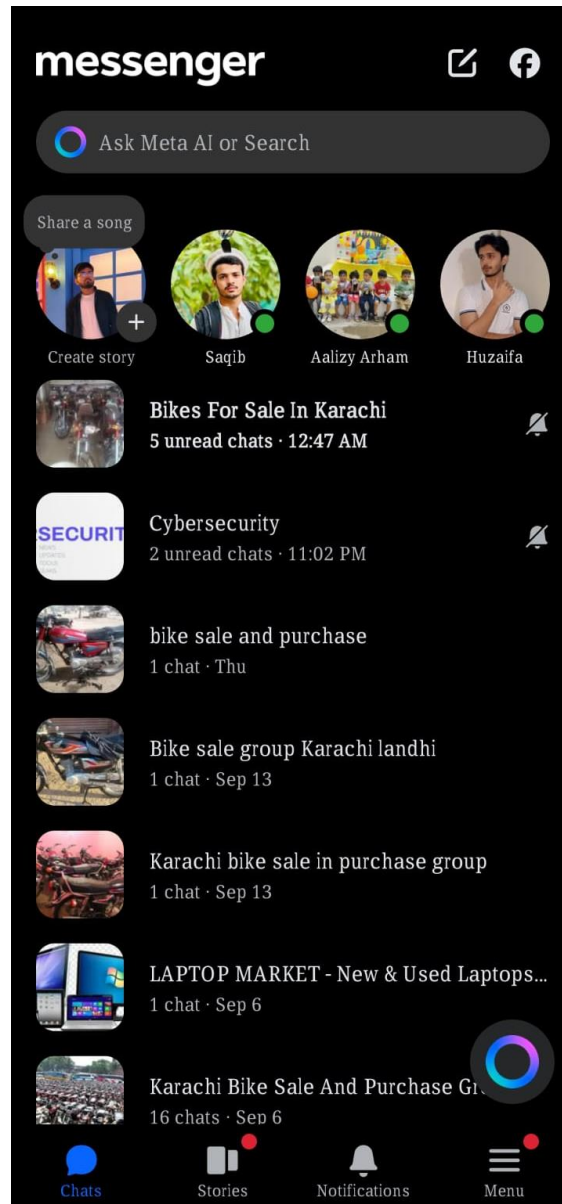


Wireframe: Page-3



Reference Images:





LAB 03

Typography & Color Theory in UI Design

Objective:

- Understand the role of typography and color in interface design.
- Learn how to add, format, and style text using different fonts.
- Explore color theory and apply it through color codes in Figma.
- Learn how to insert and mask (clip) images inside shapes for creative layouts.

Introduction:

- Typography ensures readability & tone.
- Colors communicate emotions and brand identity.
- Together, they create visual hierarchy and user engagement.

Typography Basics

- **Typography** = the art of arranging text so it's readable and visually appealing.
- **Type Tool (T)** – used to add text in Figma/other design tools.
- **Font Style** – appearance of characters (Regular, Bold, Italic, Light etc.).
- **Font Size** – height of text; measured in pixels (px) or points (pt).
- **Font Family (Types)** –
 - **Serif** (e.g., Times New Roman) – formal, traditional
 - **San Serif** (e.g., Poppins, Roboto) – clean, modern
 - **Script** (e.g., Pacifico) – handwritten, creative
 - **Display/Decorative** – used for headings/logos

In Figma

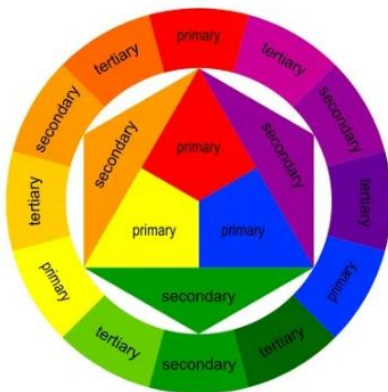
- Use Text Tool (T) → Type text → Right panel → Change Font family, Size, Weight, Line Height, and Letter Spacing.

Color Theory Basics

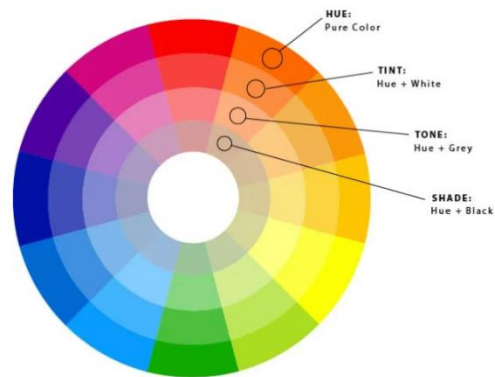
- **Primary Colors:** Red, Green, Blue
- **Secondary Colors:** Mix of two primary colors (e.g., Red + Blue = Purple).
- **Warm Colors:** Red, Orange, Yellow (energy, excitement).
- **Cool Colors:** Blue, Green, Purple (calm, trust).

Color Schemes

- **Monochromatic** – one base color + its tints/shades.
- **Analogous** – colors next to each other on the color wheel.
- **Complementary** – opposite colors on the wheel (contrast).



Types of Colors



Color Variants: Hue, Tint, Tone and Shade

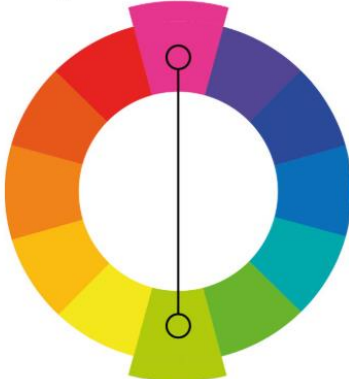
Monochromatic Color Scheme



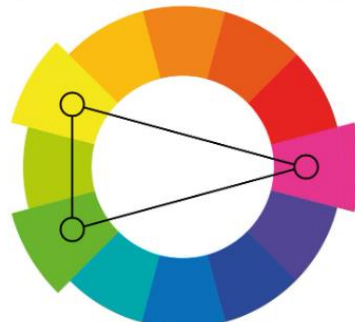
Analogous Color Scheme



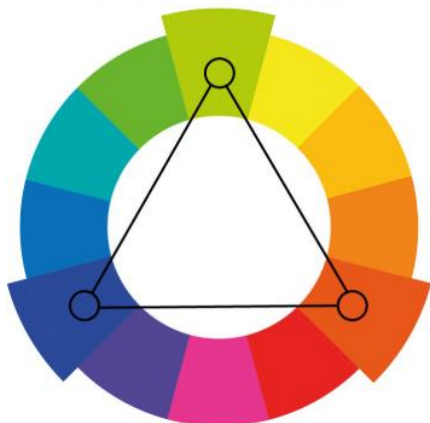
Complementary Color Scheme



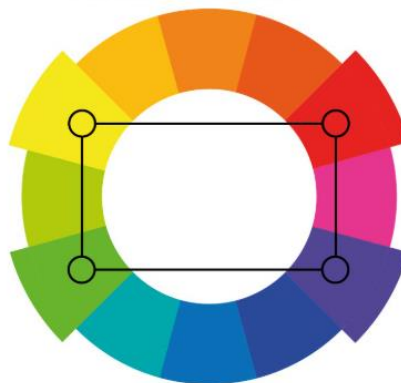
Split- Complementary Color Scheme

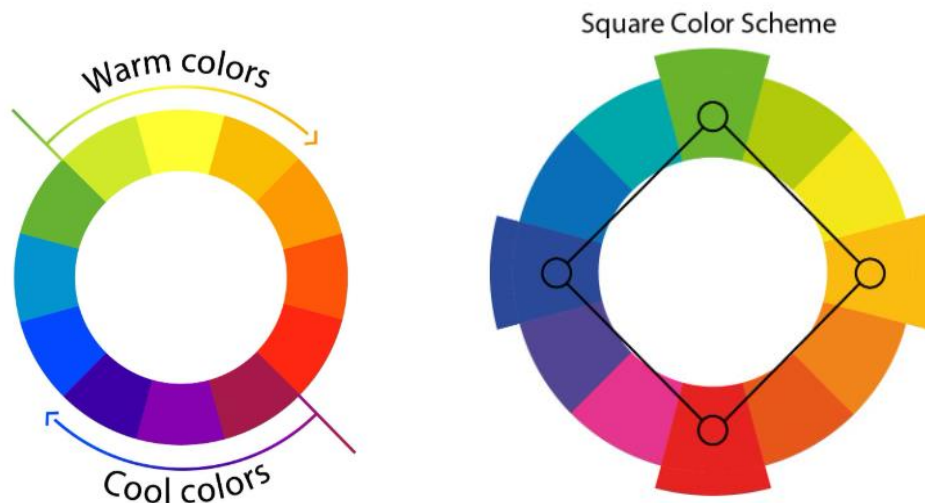


Triadic Color Scheme



Tetradic Color Scheme





In Figma

- Select any object → Right panel → Fill → click color box → enter color code (HEX/RGB).
Example: #FF5733 (orange shade).

Clipping Content / Masking Images

- **Meaning:** placing an image inside a shape (circle, rectangle, etc.).
- **Steps in Figma**
 1. Draw a shape (Ellipse or Rectangle).
 2. Import image (File → Place Image or drag from computer).
 3. Position image over shape.
 4. Select both → Right-click → **Use as Mask** (or Ctrl + Alt + M).

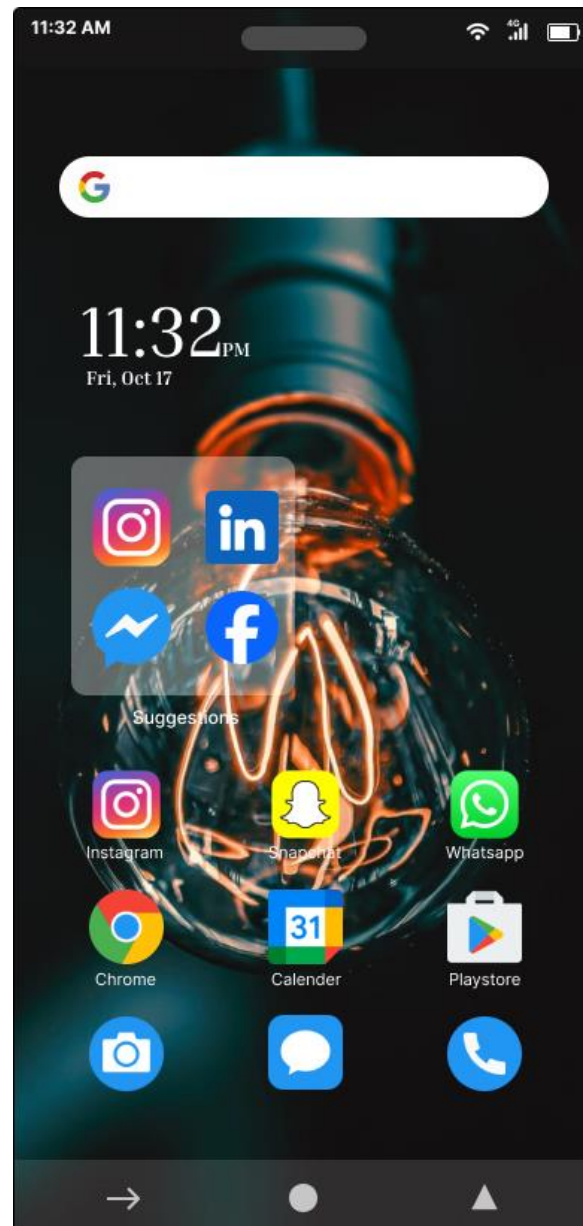
Task:

- Apply Color, Typography, image & icons on lab 2 both tasks. (Home Page Screen & Mobile Application Screen).
- Export Jpeg of frames and paste in word file.
- Copy Figma file URL of both tasks and paste in below of jpeg.

Paste screen shots of both tasks and Figma link (with anyone with link) access.

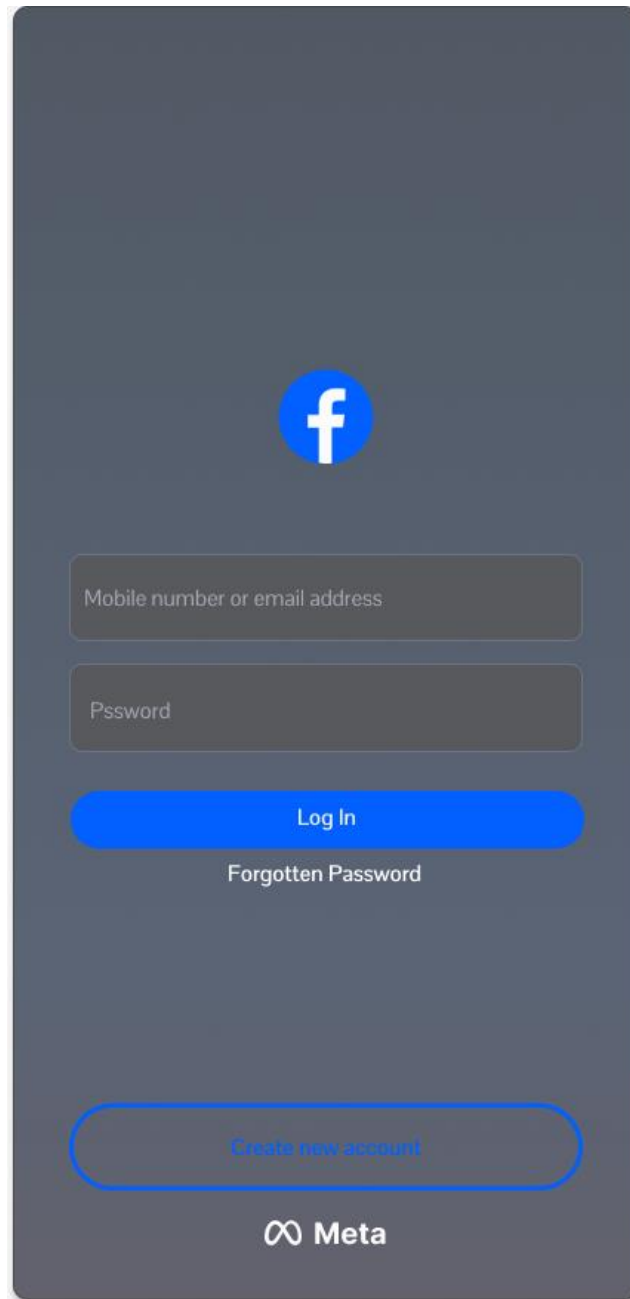
Home Screen:

Link: https://www.figma.com/design/q6wyOp4GozSTTloPAJy0QC/HCI_Lab3?node-id=41-48&t=fyYerZFiXPAVvKeg-1



Login Page:

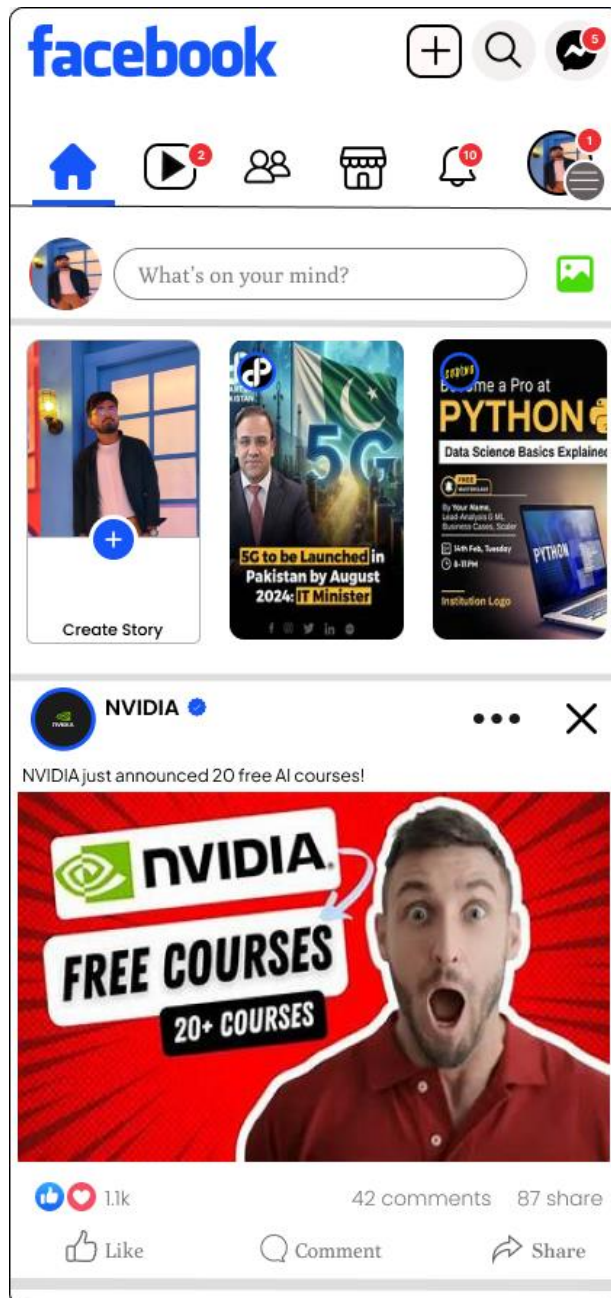
Link: https://www.figma.com/design/q6wyOp4GozSTTloPAJy0QC/HCI_Lab3?node-id=0-1&p=f&t=fyYerZFiXPVvKeg-0



A login page design for Meta, featuring a dark gray background. At the top center is the Meta logo (a blue circle with a white 'f'). Below the logo are two input fields: the first is labeled 'Mobile number or email address' and the second is labeled 'Pssword'. Below these fields is a blue 'Log In' button. Underneath the button is a link for 'Forgotten Password'. At the bottom of the form is a blue-outlined button labeled 'Create new account'. At the very bottom of the page is the Meta logo and the word 'Meta'.

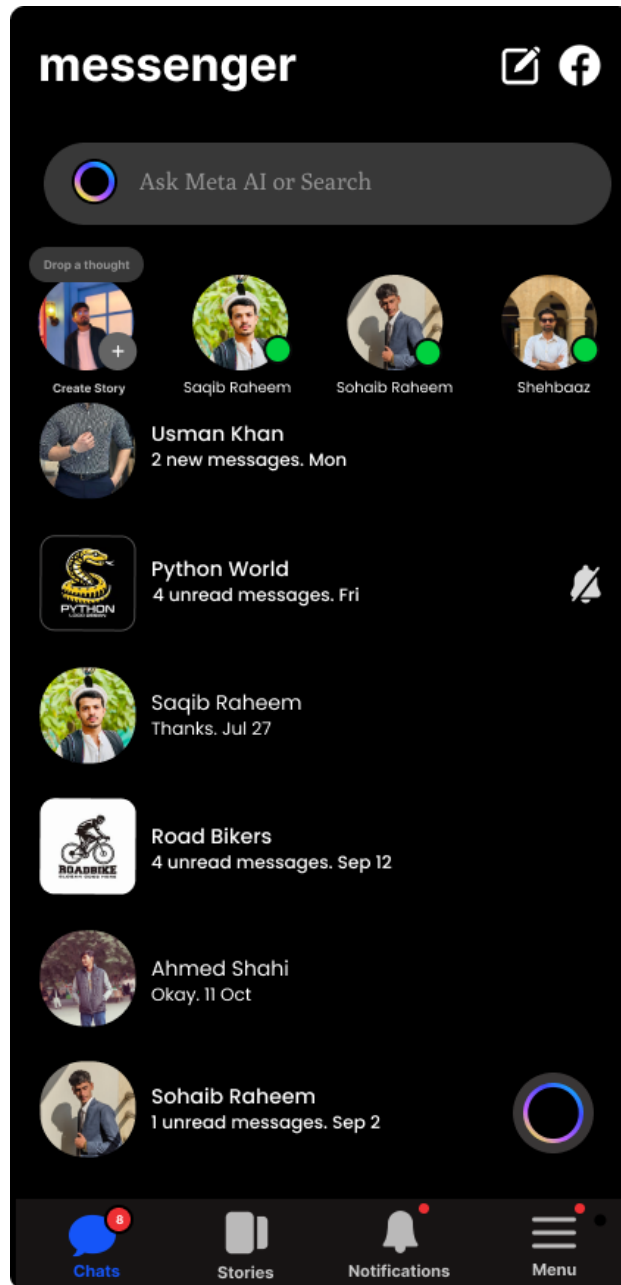
Home Page:

Link: https://www.figma.com/design/q6wyOp4GozSTTloPAJy0QC/HCI_Lab3?node-id=4-12&p=f&t=fyYerZFiXPvVvKeg-0



Messenger Page:

Link: https://www.figma.com/design/q6wyOp4GozSTTloPAJy0QC/HCI_Lab3?node-id=23-39&p=f&t=fyYerZFiXPAVvKeg-0



LAB 04

UI Elements (Forms & Buttons)

Objective:

- Understand what UI elements (like buttons, input fields, checkboxes, and labels) are.
- Learn how to design **interactive forms** with proper alignment, spacing, and feedback.
- Apply **color theory**, **typography**, and **grid systems** learned in previous labs.
- Create a **Login + Registration interface** using good design principles.

What Are UI Elements?

UI (User Interface) elements are **building blocks** of digital interfaces — they let users interact with the system.

They include:

- **Text Fields** → For input (like “Email”, “Password”)
- **Buttons** → To perform actions (Submit, Login)
- **Checkboxes** → To select multiple options
- **Radio Buttons** → To select only one option
- **Dropdowns / Combo Boxes** → To pick from a list
- **Labels** → To describe what a field does

These elements help make an app or website *usable and intuitive*.

Importance of Form Design:

Forms are the **heart of interactivity** — users log in, register, or search using forms.

A good form must be:

Simple — Only ask for necessary details.

Aligned properly — Use grid or column-based layout.

Readable — Clear labels and spacing.

Error-Friendly — Show helpful messages like “Invalid Email” or “Password too short.”

Buttons in UI

Buttons are **action triggers**.

Examples: “Submit”, “Register”, “Login”, “Cancel”.

A professional button should have:

- **Clear text** describing its purpose.
- **Consistent shape** (rounded corners look friendly).
- **Hover State** → Slight color change when hovered.
- **Active State** → Darker tone when clicked.
- **Disabled State** → Greyed out, non-clickable look.

Example:

Button State	Description
Default	Blue “Login” button
Hover	Lighter blue
Active	Darker blue
Disabled	Grey

Consistency & Feedback

- **Consistency:** Use same fonts, button sizes, and colors across all screens.
- **Feedback:** Inform users what happened (e.g., “Login Successful!” or “Incorrect Password”).

Radio Button vs Checkbox

Type	Use Case	Behavior
Radio Button	When user can select only one option	Selecting one deselects others
Checkbox	When user can select multiple options	Can check/uncheck independently

Example:

- Gender → Radio Buttons (Male, Female, Other)
- Interests → Checkboxes (Reading, Sports, Music)

Real-World Examples

App	Key Elements
Instagram Login	Simple form, primary blue button, feedback for wrong password
Google Sign-In	Clean layout, minimal colors, consistent spacing

Figma Components & Variants

What are Components?

- A **component** in Figma is a reusable design element (e.g., button, input field).
- You can make one component and reuse it everywhere.

Variants in Components

Variants allow you to make **different states** of the same component (like Default, Hover, Active, Disabled).

You can switch between them easily.

Example:

- Create one “Button” component.
- Make variants:
 - Default (Blue)
 - Hover (Light Blue)

- Pressed (Dark Blue)
- Disabled (Grey)

Task 1: Design a Login Form

Elements to include:

- Text fields: “Email”, “Password”
- Primary Button: “Login”
- Secondary Text: “Forgot Password?”
- Checkbox: “Remember Me”

Requirements:

- Apply color theory (use brand colors)
- Use consistent spacing with grid alignment
- Add hover and pressed state for button

Task 2: Design a Registration Form

Elements to include:

- Text fields: Name, Email, Password, Confirm Password
- Radio Buttons: Gender (Male/Female/Other)
- Checkbox: Accept Terms & Conditions
- Primary Button: “Register”

Requirements:

- Follow same font and color style as Login form
- Use feedback text under input boxes
- Maintain proper alignment using grid

Paste screen shots of both tasks and Figma link (with anyone with link) access.

Task 1:

Figma Login Form Link:

<https://www.figma.com/design/880cOANr2xOnNvykIzUcY5/HCI-Lab-4?node-id=4-74&p=f&t=cvvdDVBjVGtOFzOD-0>

- Login Form with components & variants.

The image displays a design system for a login form, divided into two main sections: a component library on the left and a final assembled form on the right.

Component Library (Left):

- A dashed purple box contains two "Login" buttons. The top button is bright blue with white text, and the bottom button is a darker blue with white text.
- Below these buttons, another dashed purple box contains two checkboxes. The top checkbox is an empty square, and the bottom checkbox is a square with a white checkmark on a blue background.

Assembled Form (Right):

- The title "Login" is displayed in a large, bold, black font.
- Below the title, the label "Email" is followed by a light gray rectangular input field.
- Below the email field, the label "Password" is followed by another light gray rectangular input field.
- Below the password field, there is a checkbox (an empty square) followed by the text "Remember Me".
- Below the "Remember Me" text is a large, bright blue button with the word "Login" in white text.
- At the bottom of the form is a link that says "Forgot Password?" in a blue font.

Task 2:

Figma Registration Form Link: <https://www.figma.com/design/880cOANr2xOnNvykIzUcY5/HCI-Lab-4?node-id=0-1&p=f&t=cvvdDVBJVGtOFzOD-0>

- Registration Form with Components & variants.

Registration Form

Name

Email

Password

Confirm Password

☐ Male ☐ Female ☐ Other

☐ Accept Terms & Conditions

Register

LAB 05

Slider Menu Bar & Scrolling Behaviour

Objective

This lab will teach students how to design a slider (side) menu bar in Figma and apply proper slide-in interactions. Students will also learn how to use vertical and horizontal scrolling for long and wide content. The aim is to build a clean one-page mobile app interface with a functional menu bar. Students will practice using frames, prototype settings, and scrolling behaviors effectively.

1. What is a Slider Menu Bar?

A **Slider Menu Bar** (also called Side Menu or Drawer Menu) is a hidden navigation panel that slides in from the left or right side of the screen when the user taps the **menu icon (hamburger icon)**.

Why do we use a Slider Menu Bar?

- Saves screen space
- Shows only when needed
- Makes navigation easy in mobile apps
- Looks modern and user-friendly

2. Steps to Create a Slider Menu Bar in Figma

Step 1: Create Main Screen

- Create a **Mobile Frame** (e.g., iPhone 14).
- Add header → Add a **menu (≡)** icon on the top left.

Step 2: Create the Side Menu Panel

- Create a new **Frame** (same height as mobile).
- Give background color (white/gray).
- Add menu items such as:

- Home
- Profile
- Settings
- Help
- Place this frame **outside the mobile frame** on the left side.

Step 3: Overlay for Slide Effect

- Create a rectangle with low opacity (e.g., 30% black) to show background dimming when menu opens.

Step 4: Add Interaction

- Select the **menu icon** → **Prototype** tab
- Interaction:
 - **On Tap** → **Open Overlay** → **Menu Open Frame**
 - Animation: **Slide In Left**

3. Scrolling Behaviour in Figma

Figma gives two types of scroll:

A) Vertical Scrolling

Vertical scrolling means the user can scroll **up–down**.

Where Vertical Scroll is Used?

- Long pages
- Articles / Blogs
- Lists (Contacts, Messages, Menu)
- Forms
- Dashboards with long content

How to Apply Vertical Scroll?

1. Select your **Mobile Frame**
2. On the right panel → **Prototype**
3. Under **Overflow Behaviour** → **Scroll Vertically**
4. Make your content longer than the screen height

B) Horizontal Scrolling

Horizontal scrolling means content moves **left–right**.

Where Horizontal Scroll is Used?

- Image galleries
- Product sliders
- Category sliders
- Story feature (like Instagram stories)
- Horizontal card lists

How to Apply Horizontal Scroll?

1. Create a **Group** or **Frame** wider than the screen width
2. Select the group/frame
3. Right side → **Clip Content: ON**
4. Prototype → Set **Horizontal Scroll**

4. Overall, What Students Learned

- How to build a complete **slider side menu**
- How to apply **prototype linking with slide animation**
- How to create **overlay background**
- How to apply **vertical and horizontal scrolling**
- How to organize screens for clear interactions

Task 01: One-Page App with Slider Menu Bar

Students must create:

1. One mobile frame
2. A proper menu bar frame
3. Screenshot 1 → Main screen with menu icon
4. Screenshot 2 → Menu bar frame separately
5. Screenshot 3 → Menu opened on same screen
6. Screenshot 4 → Interaction panel screenshot (settings visible)

Requirements:

- Include **at least 4–5 menu items**
- Use **Slide In Left** animation
- Add **overlay background**
- Neat alignment & spacing required

Task 02: Apply Vertical + Horizontal Scroll on the Same Page

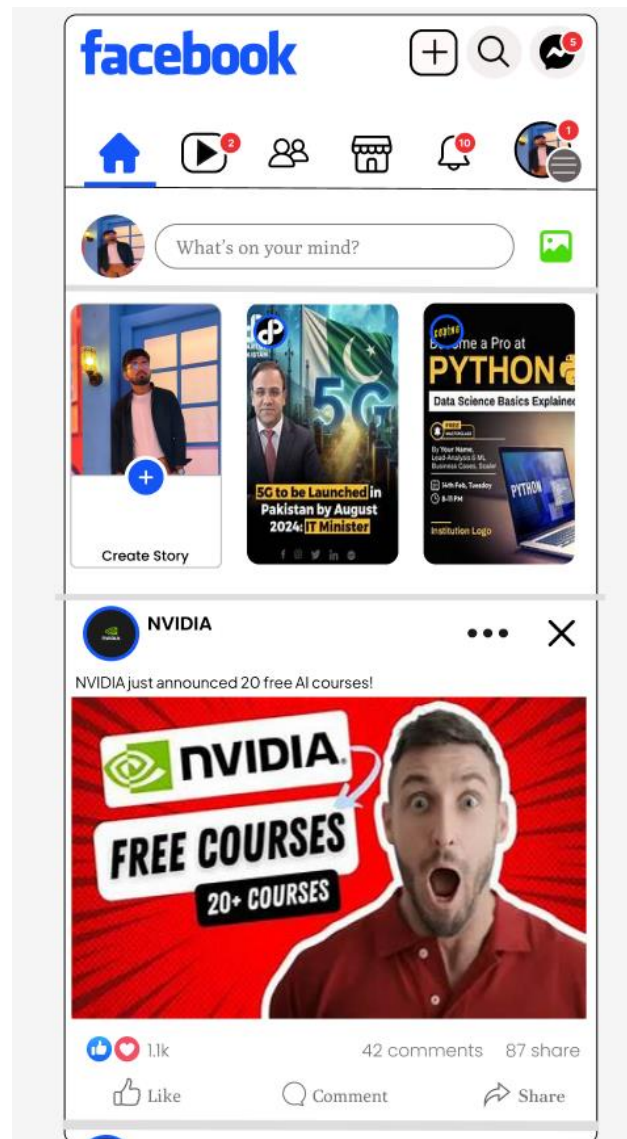
Students must:

1. Create a **mobile screen**
2. Upper area: **Horizontal scroll** (image slider or category cards)
3. Lower area: **Vertical scroll** (long content or list)
4. Proper screenshots:
 - Frame with vertical scroll
 - Frame with horizontal scroll
 - Final prototype running

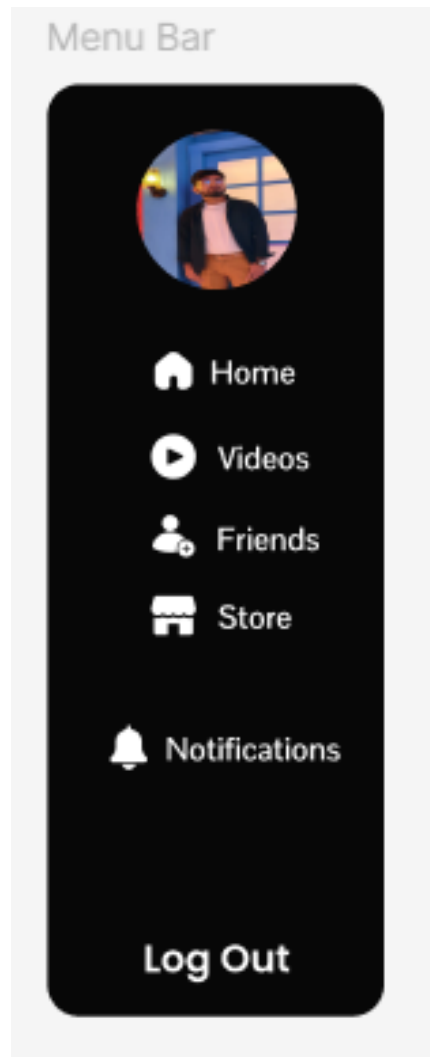
Paste screen shots of both tasks and Figma link (with anyone with link) access.

Task 1:

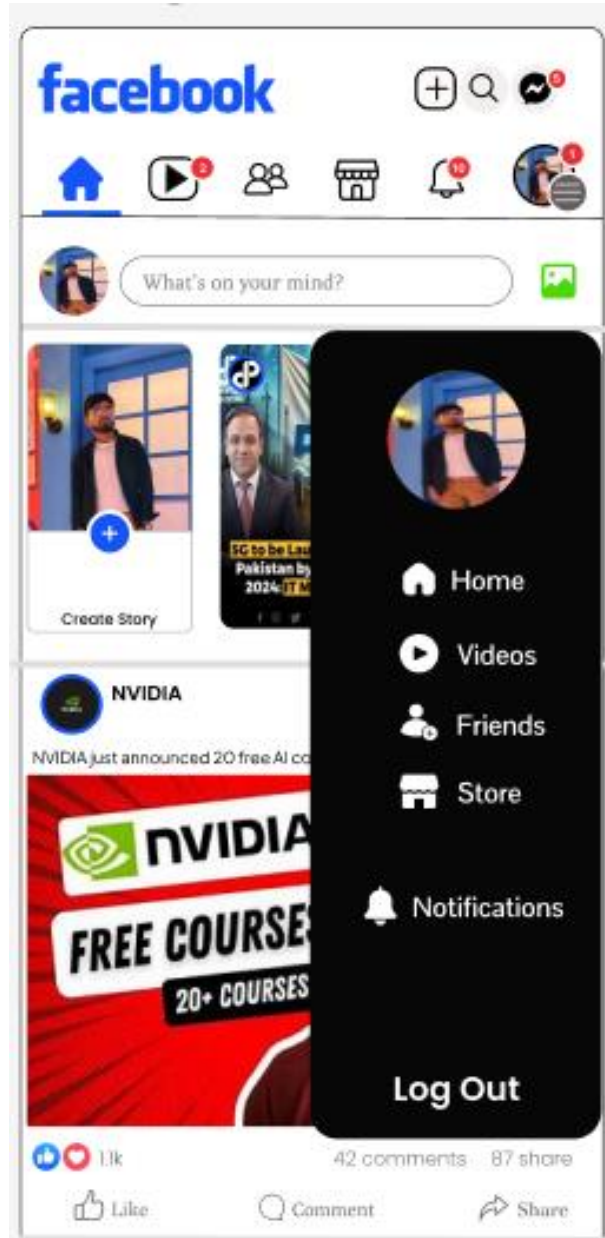
Screenshot 1:



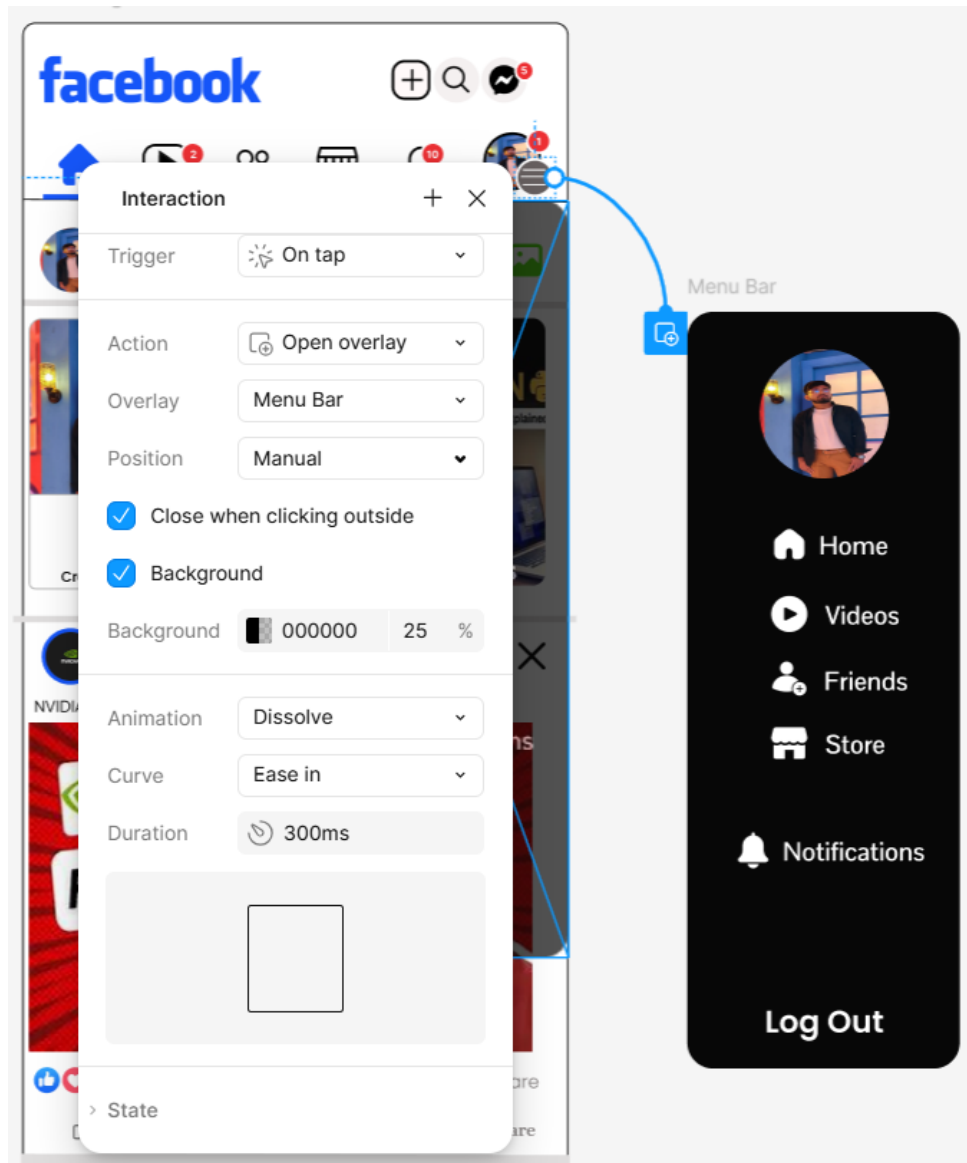
Screenshot 2:



Screenshot 3

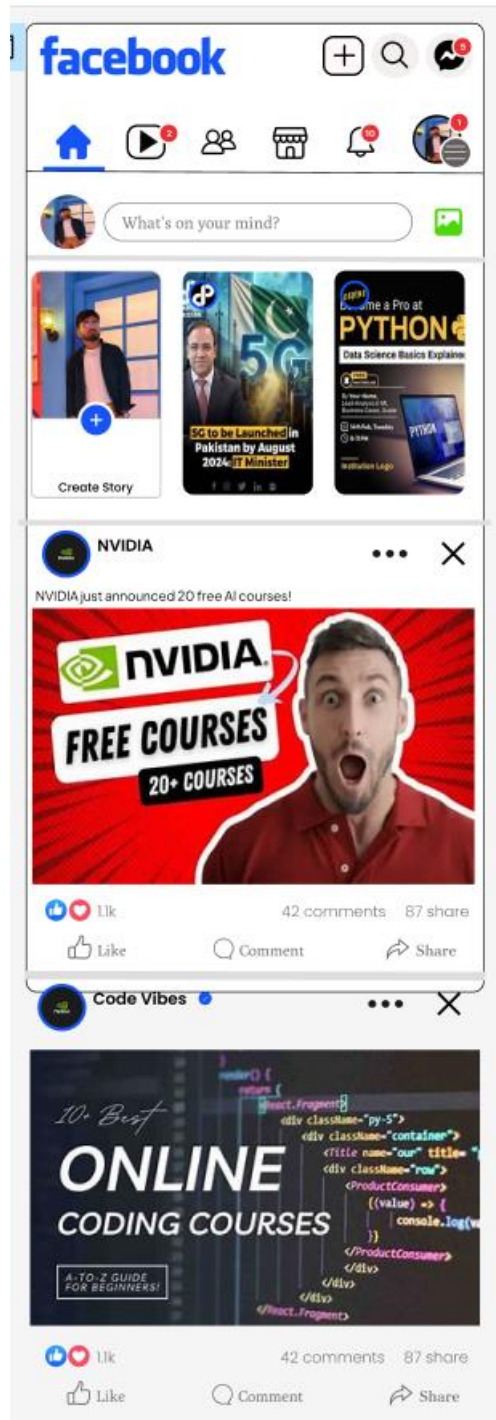


Screenshot 4



Task 2:

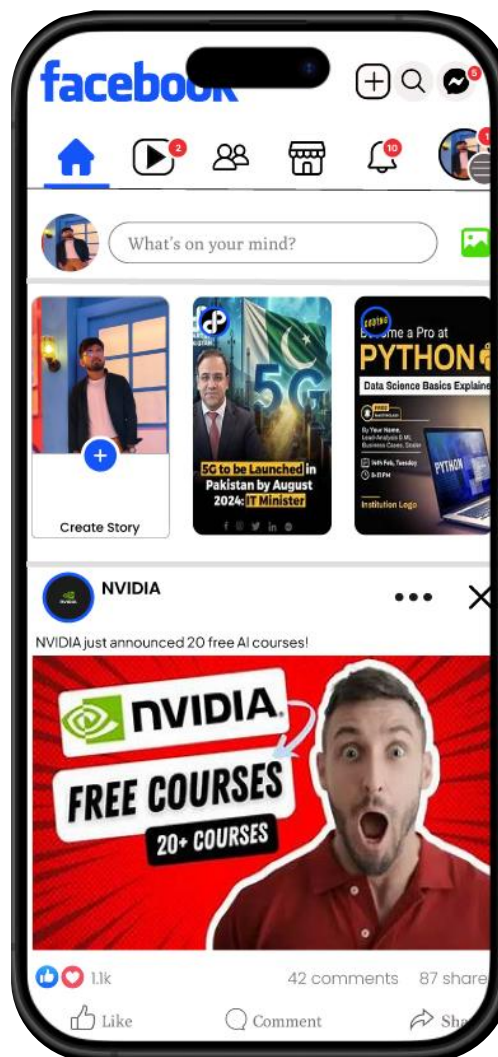
Vertical Scroll:



Horizontal Scrolling:



Final Prototype:



LAB 06

Interaction between Screens by using Prototyping

Objective:

This lab will help students create a **multi-screen app prototype** with complete interaction flow. Students will apply all techniques learned so far, including **hover effects, checkboxes, menu bars, scrolling, and footers**.

The goal is to build a **4–5-page app** with smooth navigation between screens. Students will also learn to show **prototype wiring and interactions** through screenshots.

LAB TASK (Final Submission)

Task: Build a 5-page Mobile App Prototype

You must design a **5-screen app** following the topics covered so far.

Your app must include:

Mandatory Features

- Signup → Login → Home → About → Services
- Working **side menu bar**
- **Hover effects** on buttons
- **Check box** on signup
- **Vertical OR Horizontal scrolling**
- **Footer** (simple or animated)
- **Automatic Carousel** on images.
- All screens must be linked properly
- Show full **prototype wiring**

Deliverables

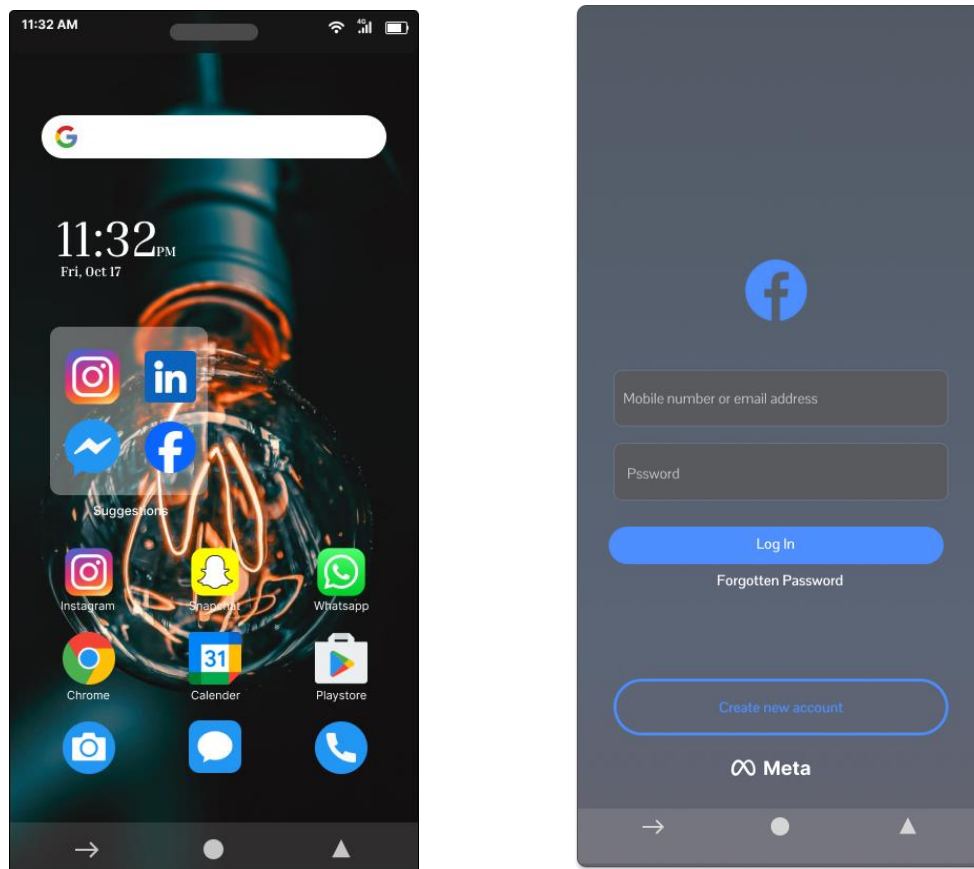
- All 5 screen designs
- Screenshot of menu bar
- Screenshot of scrolling setup
- Screenshot of interaction settings
- Final prototype wiring screenshot
- Paste screen shots of tasks and Figma link (with anyone with link) access.

Lab Task:

Link: https://www.figma.com/design/q6wyOp4GozSTTioPAJy0QC/HCI_Lab3?node-id=0-1&p=f&t=5ySH2rMdIlJsLhTb-0

- In this lab task, I designed five screens in Figma: one mobile GUI screen and four Facebook-related screens, including Login, Sign Up, Home Page, and Messenger.

Figure 1: Overview of All Five Screen Designs



Sign Up

Name

Email

Password

Confirm Password

☐ Male
 ☐ Female
 ☐ Other

☐ Accept Terms & Conditions

Sign up

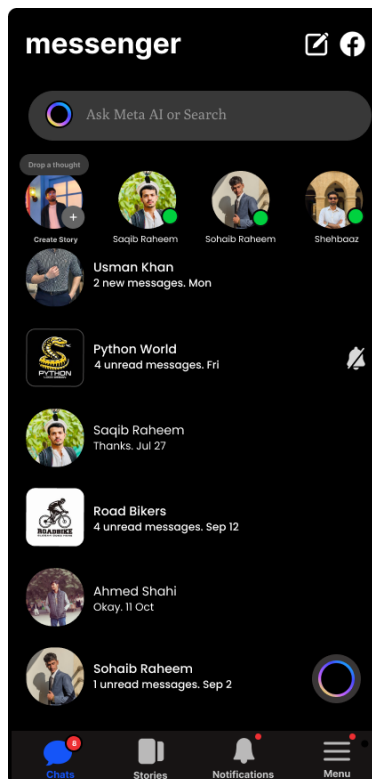
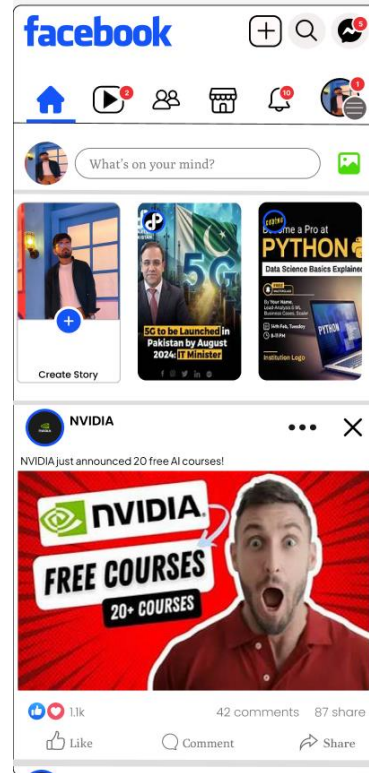


Figure 2: Menu Bar Design Screenshot

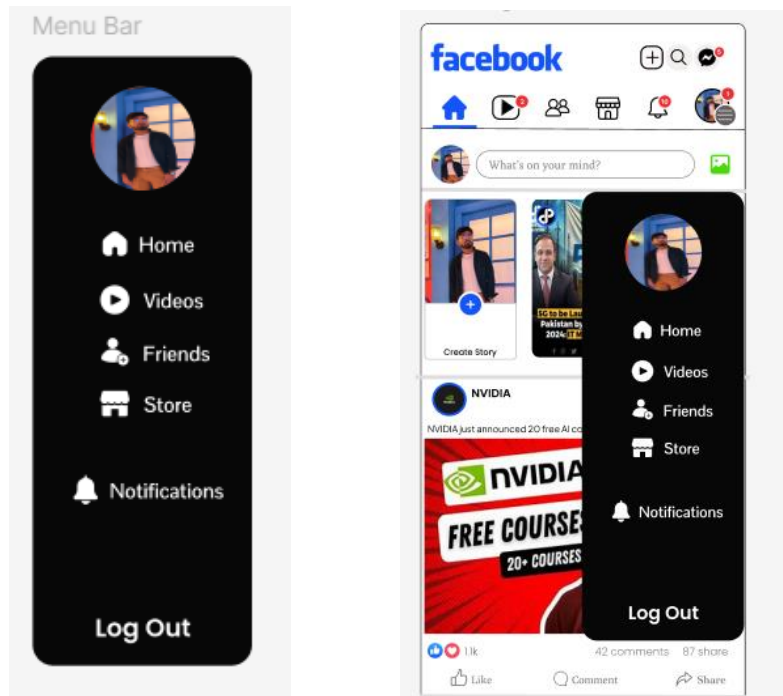


Figure 3: Scrolling Setup Configuration

Screenshots of Horizontal and vertical scrolling setup.

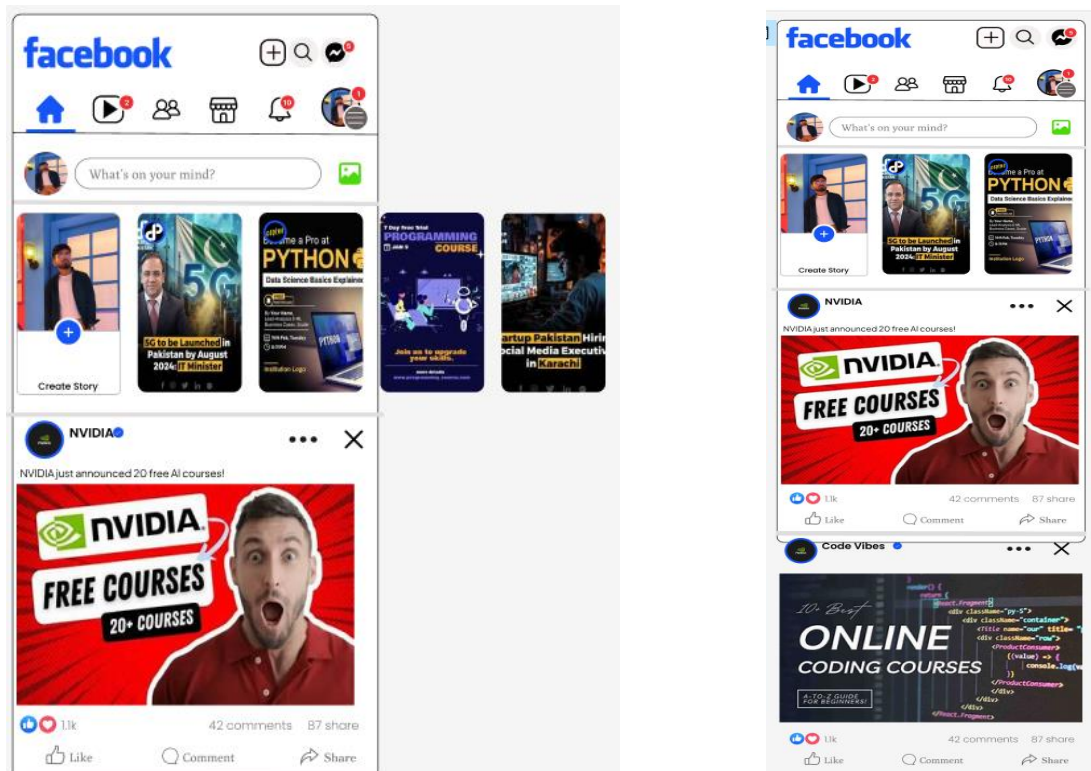
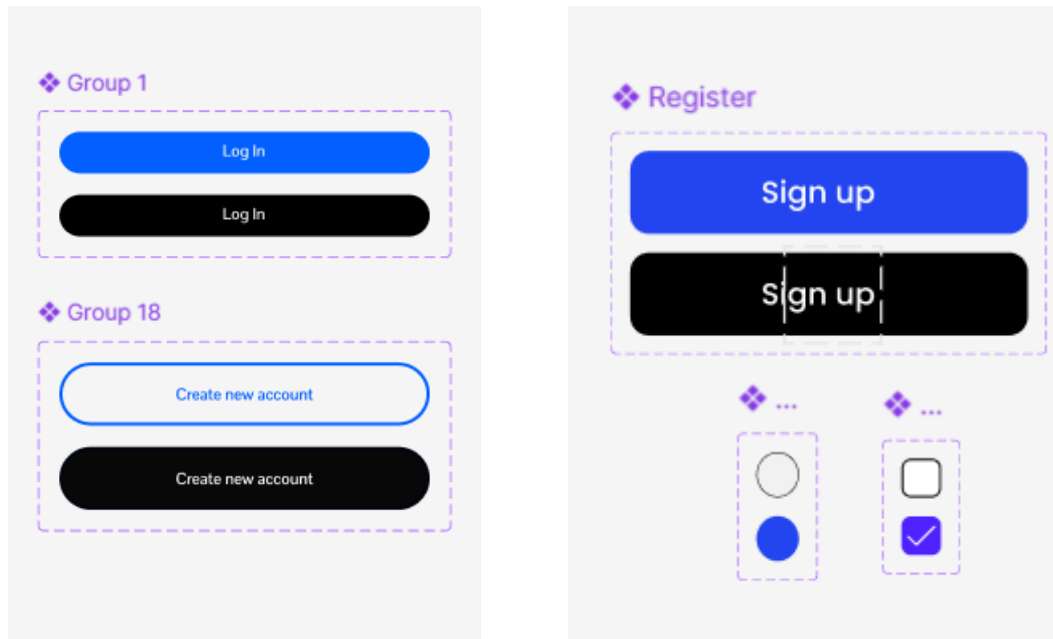


Figure 3.1: Components and variants

This component includes hover effects for the **Login** and **Create Account** buttons, **radio button** and **check boxes**.



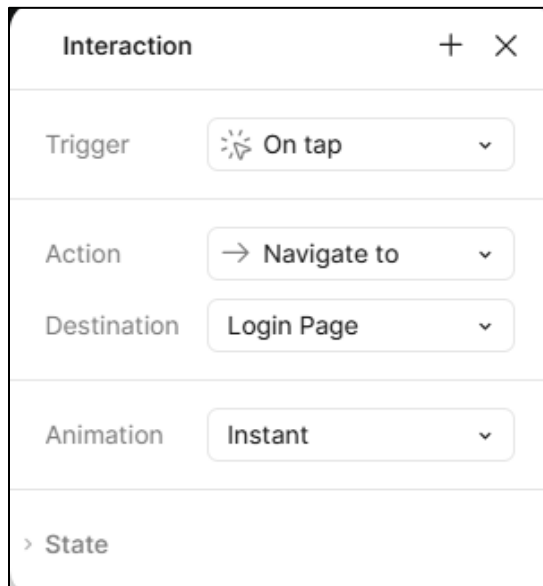
Automatic Carousel:

This component displays images in an automatic sliding carousel using Figma components.

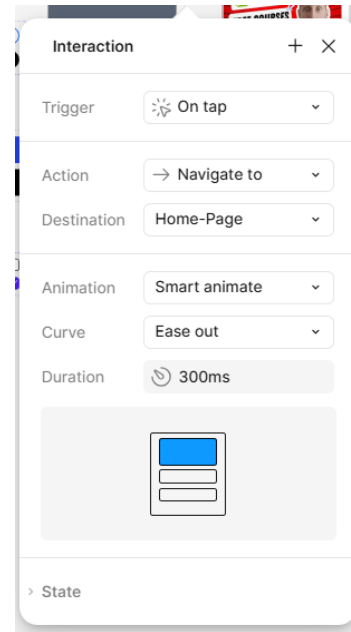


Figure 4: Interaction Settings Screenshot

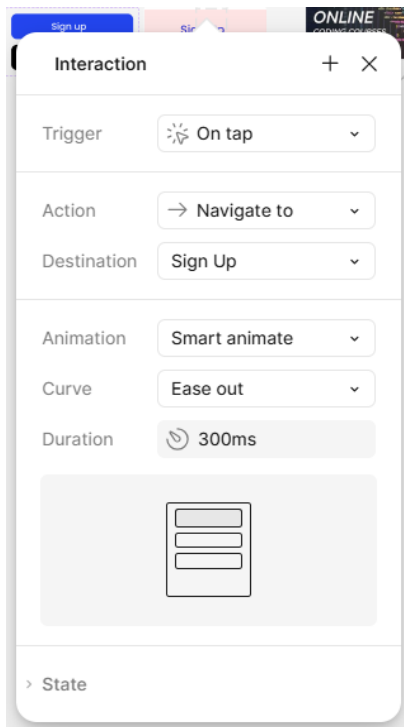
Facebook Icon → Login Screen Interaction



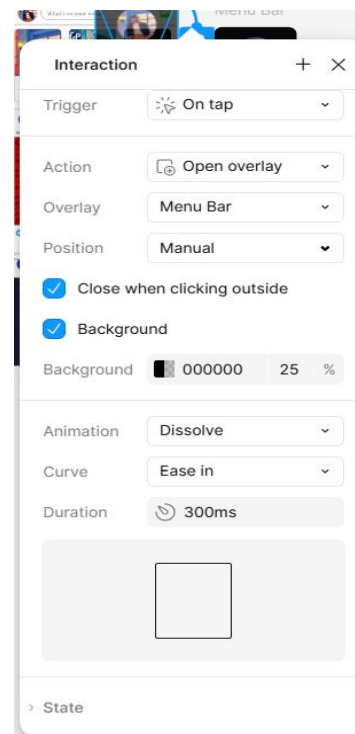
Login → Facebook home-page



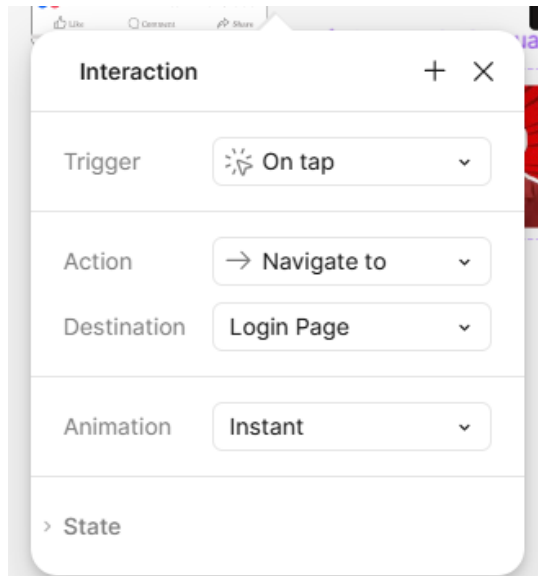
Create Account → Sign Up Interaction



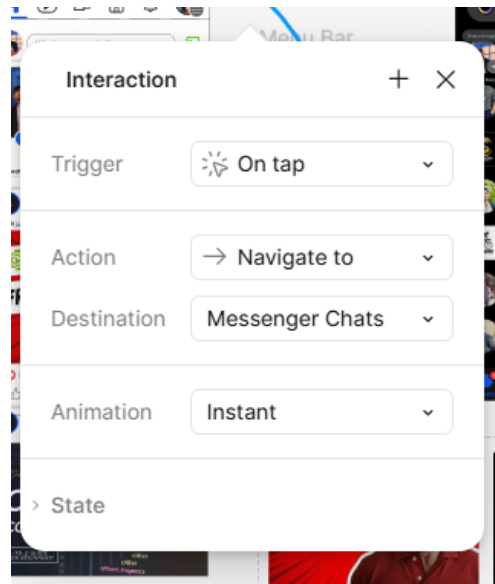
Hamburger → Menu-bar



Menu bar (logout) → Login screen



Message icon → Messenger



Login screen footer → Mobile home-screen

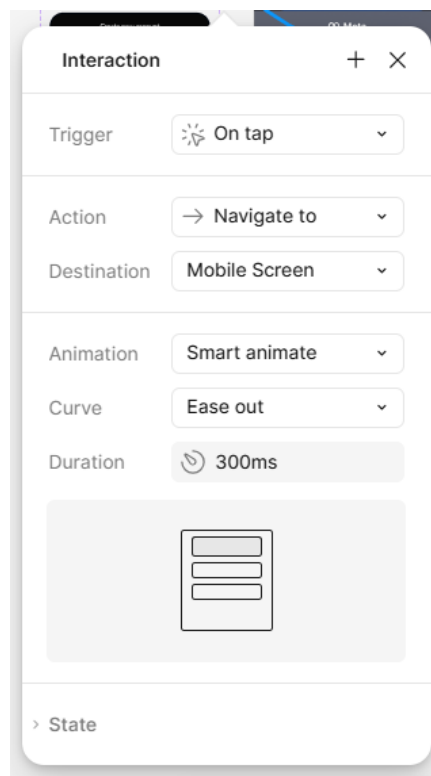
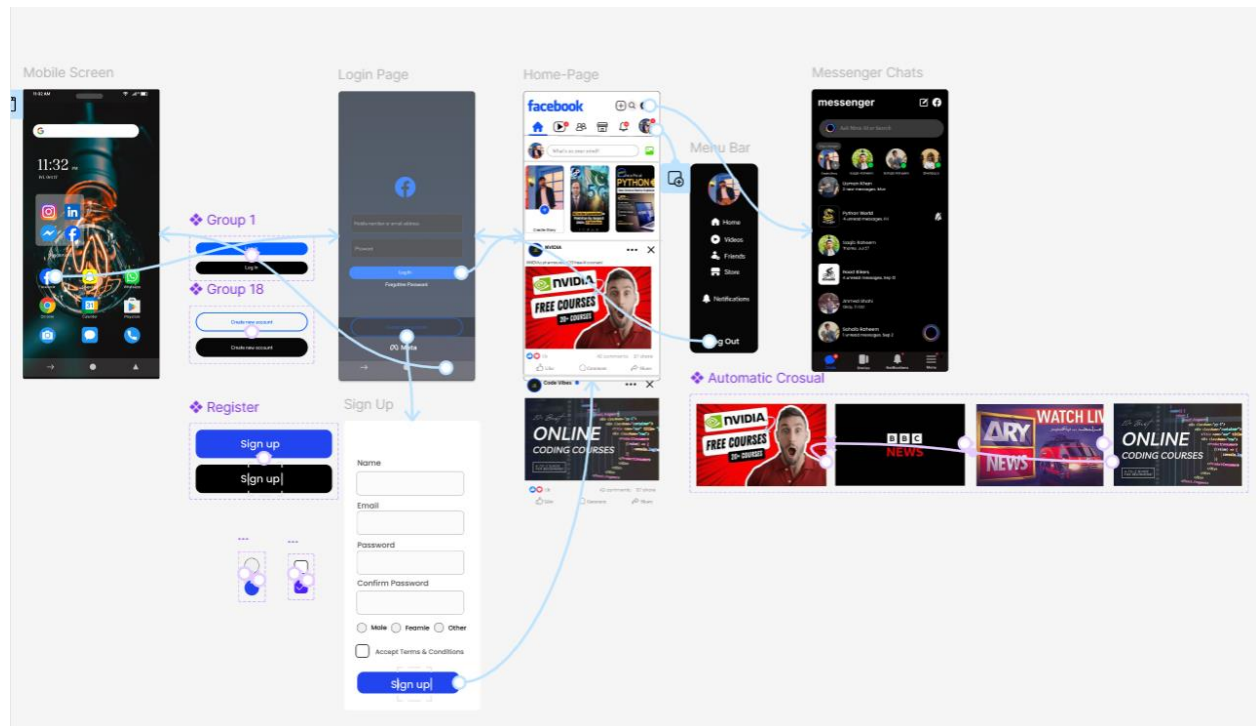


Figure 5: Final Prototype Wiring View



Figma Link All pages Link:

https://www.figma.com/design/q6wyOp4GozSTTioPAJy0QC/HCI_Lab3?node-id=0-1&p=f&t=pr4YgIoOKdcQ9knd-0

LAB 07

Introduction to Blender & Installation

Objective:

By the end of this lab, students will be able to:

- Understand what Blender is
- Know where and why Blender is used
- Install Blender successfully on their systems
- Capture installation screenshots as proof

Introduction to Blender:

Blender is a **free and open-source 3D creation software**.

It is used to create:

- 3D Models
- Animations
- Games assets
- Visual effects
- Short movies

Blender supports:

- Modeling
- Texturing
- Rendering
- Animation

Why Use Blender:

- Free (no license cost)
- Used by professionals and beginners
- Supports industry-level 3D work
- Works on Windows, macOS, Linux

Where Blender is Used:

- Game Development
- Animation & Movies
- Product Design
- Architecture Visualization
- Education & Learning

System Requirements (Basic):

- OS: Windows 10 or later
- RAM: Minimum 8GB (recommended)
- Mouse with scroll wheel
- Internet connection

Blender Installation Steps:

1. Open browser → Go to **www.blender.org**
2. Click **Download Blender**
3. Select your Operating System (Windows)
4. Download setup file
5. Open downloaded file
6. Click **Next**
7. Accept License Agreement

8. Choose installation location
9. Click **Install**
10. Click **Finish**
11. Open Blender to verify installation

Tasks:

Task 1

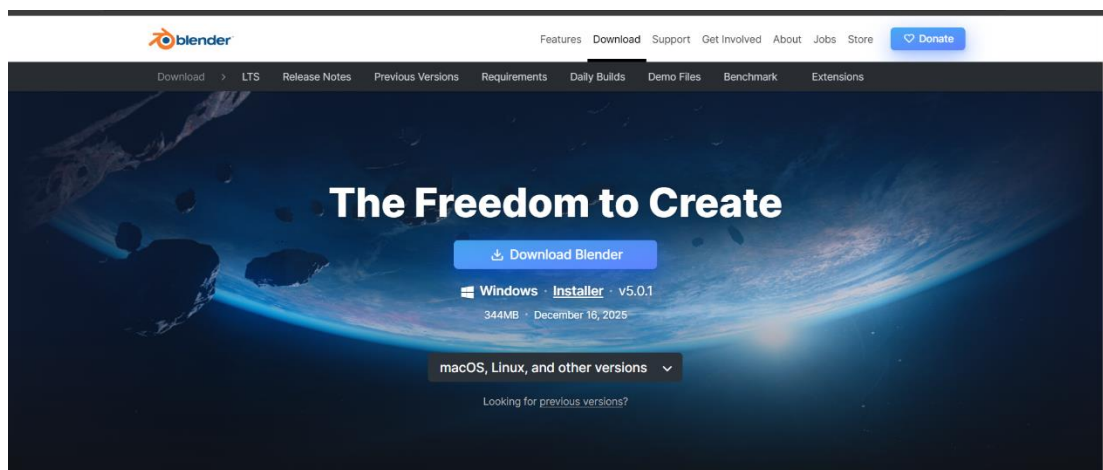
- Download and install Blender on your system

Task 2

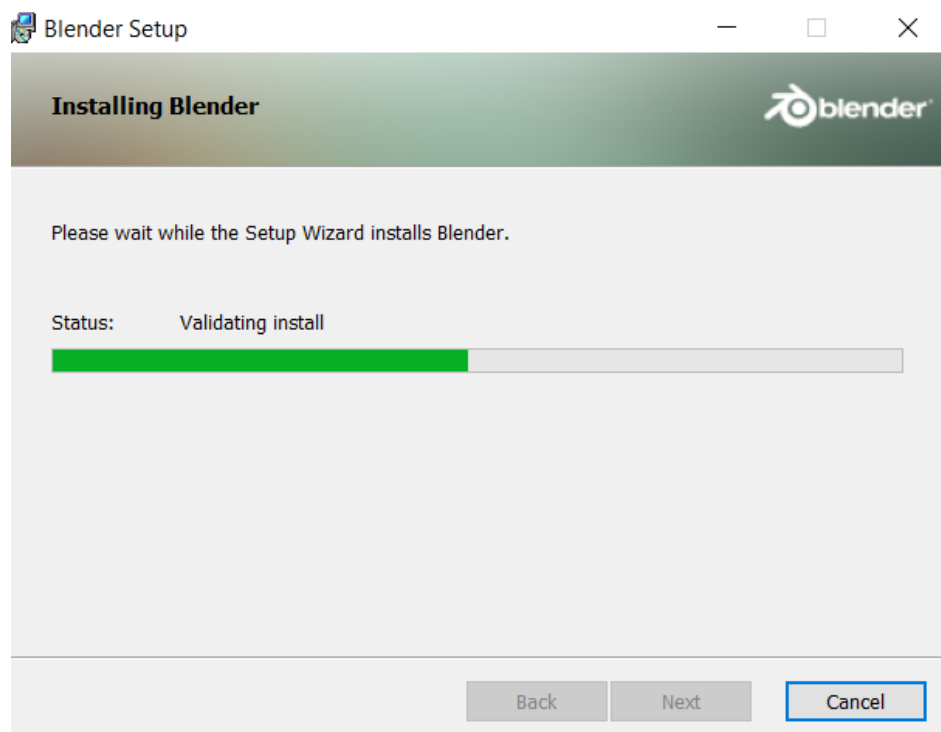
- Take screenshots of:
 - Blender download page
 - Installation process
 - Blender opening screen

Lab Task:

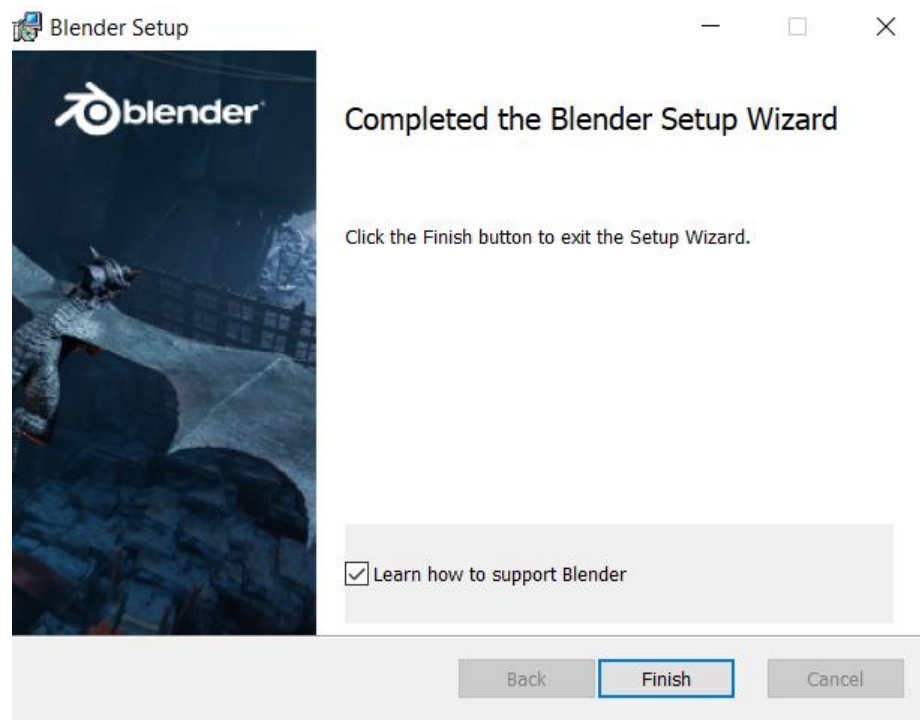
Open the Blender official website and see the download for Windows 10



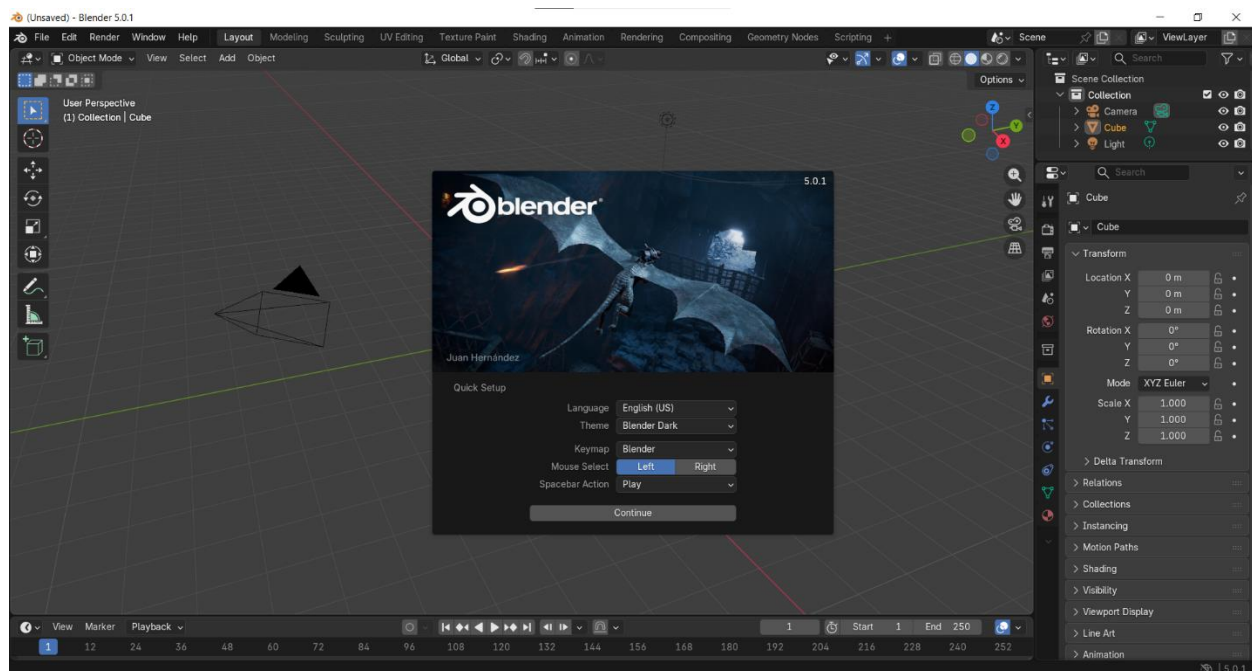
Download the setup and run the .exe file with admin privileges.



After that, click on finish, and Blender has successfully installed



Blender Interface



LAB 08

Blender Basics (Object Mode & Mesh)

Objective:

Students will learn:

- What Mesh is
- How to add objects
- Object Mode tools
- Move, Scale, Rotate using X, Y, Z axis
- Shortcut keys in Blender

What is a Mesh:

A Mesh is a **3D object made of vertices, edges, and faces.**

Examples:

- Cube
- Sphere
- Cylinder
- Plane

Add Mesh in Blender:

- Shortcut key: **Shift + A**
- Menu opens → Mesh
- Available Mesh Objects:
 - Cube
 - UV Sphere
 - Cylinder
 - Cone

- Plane
- Torus

Object Mode:

Object Mode is used to:

- Move objects
- Rotate objects
- Scale objects
- Delete objects

(Default mode when Blender opens)

Transform Tools:

Move Tool:

- Shortcut: **G**
- Axis control:
 - GX → Move on X-axis
 - GY → Move on Y-axis
 - GZ → Move on Z-axis

Scale Tool:

- Shortcut: **S**
- Axis control:
 - SX → Scale on X-axis
 - SY → Scale on Y-axis
 - SZ → Scale on Z-axis

Rotate Tool:

- **Shortcut: R**
- **Axis control:**
 - RX → Rotate on X-axis
 - RY → Rotate on Y-axis
 - RZ → Rotate on Z-axis

Other Object Mode Tools:

- **Add Cube** → Shift + A → Mesh → Cube
- **Delete Object** → Select object → Press **Delete**
- **Measure Tool** → Measure distance between objects
- **Transform Panel** → N key (Location, Rotation, Scale)

Tasks:

Task 1

- Add an Cube and move it using GX, GY, GZ

Task 2

- Scale the cube using SX, SY, SZ

Task 3

- Rotate the cube on all axes (RX, RY, RZ)

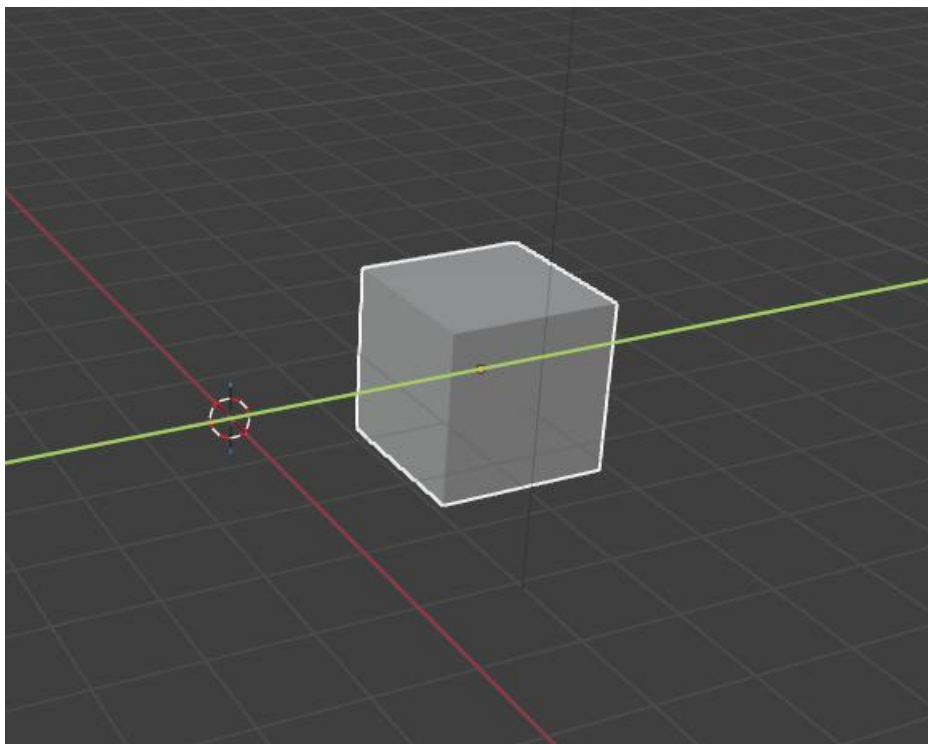
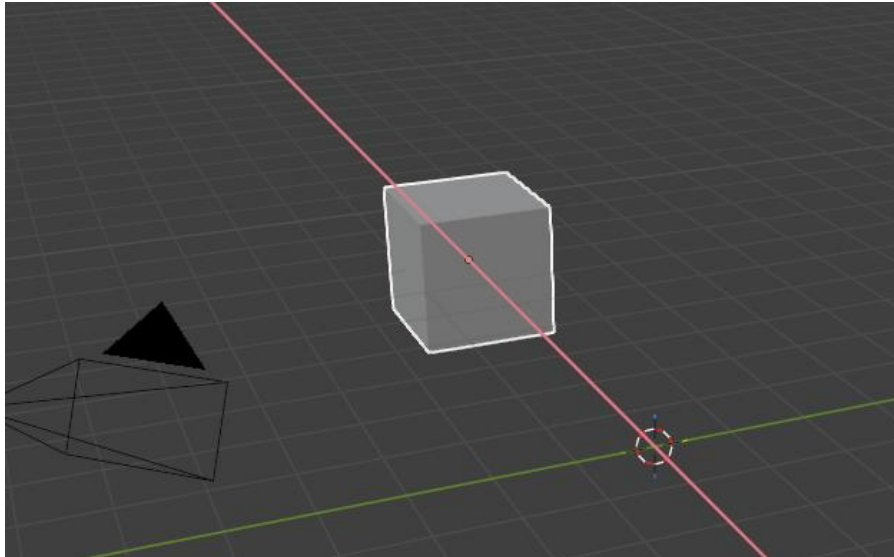
Task 4

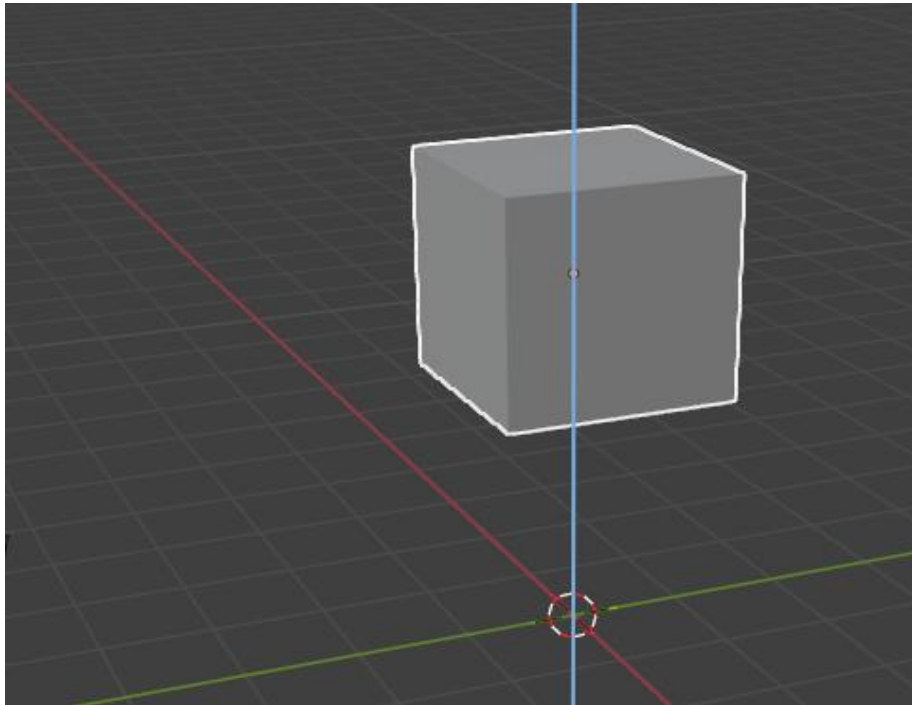
- Add another mesh object and measure distance

Take screenshots of all steps.

Task 1:

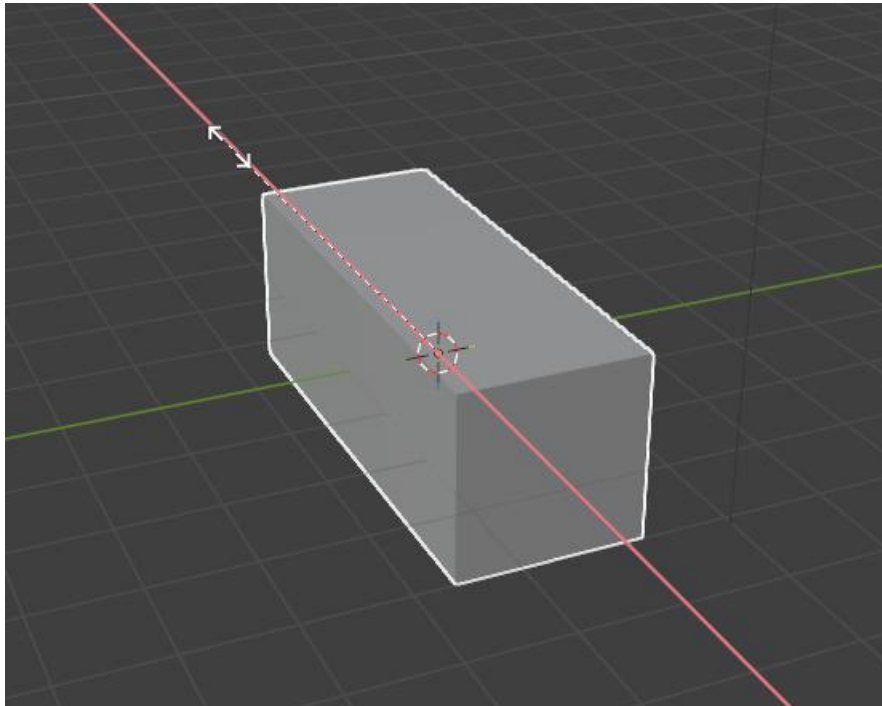
Add a Cube and move it using GX, GY, GZ

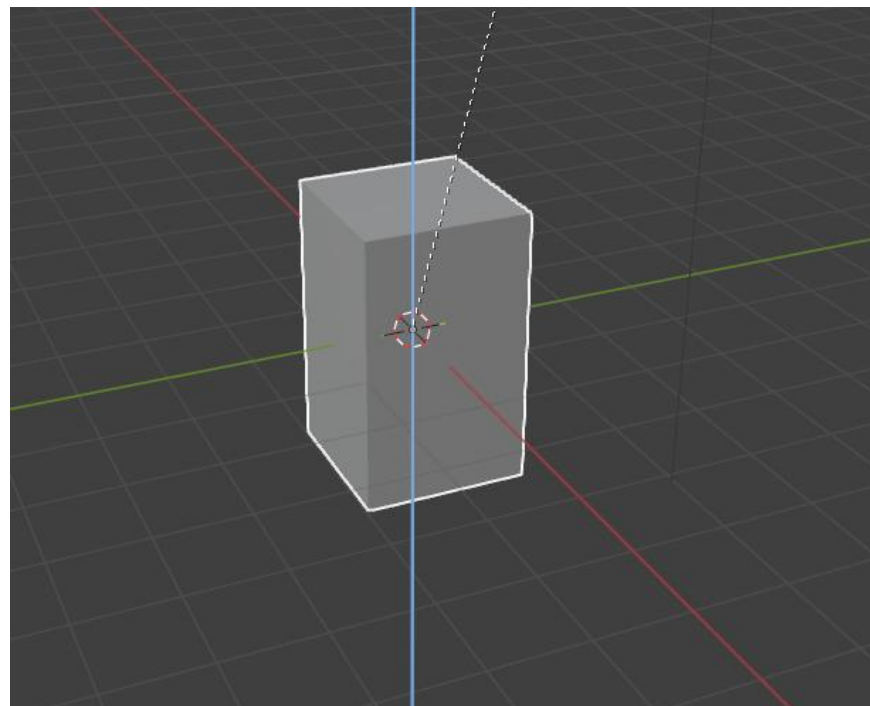
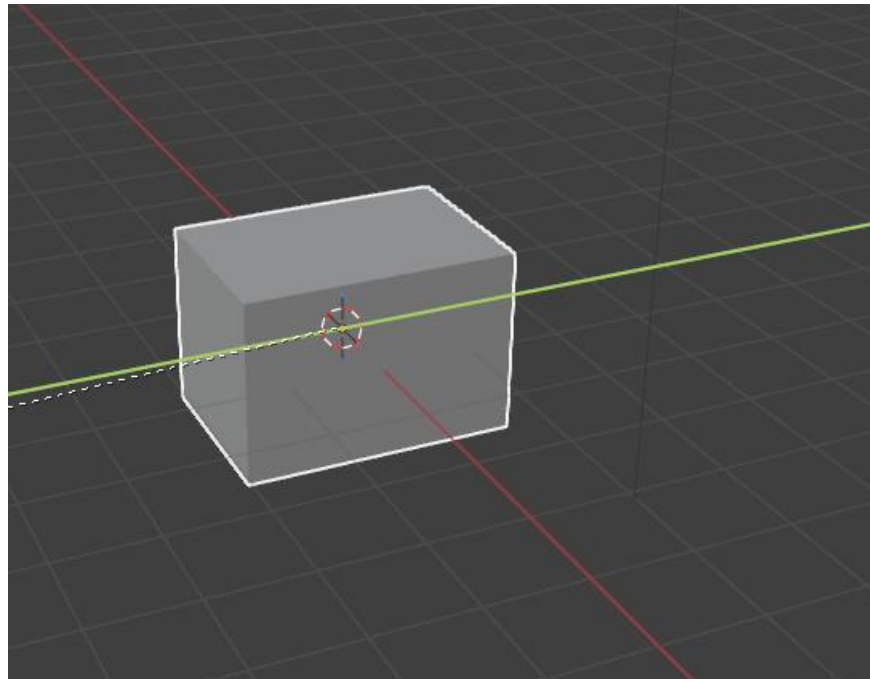




Task 2:

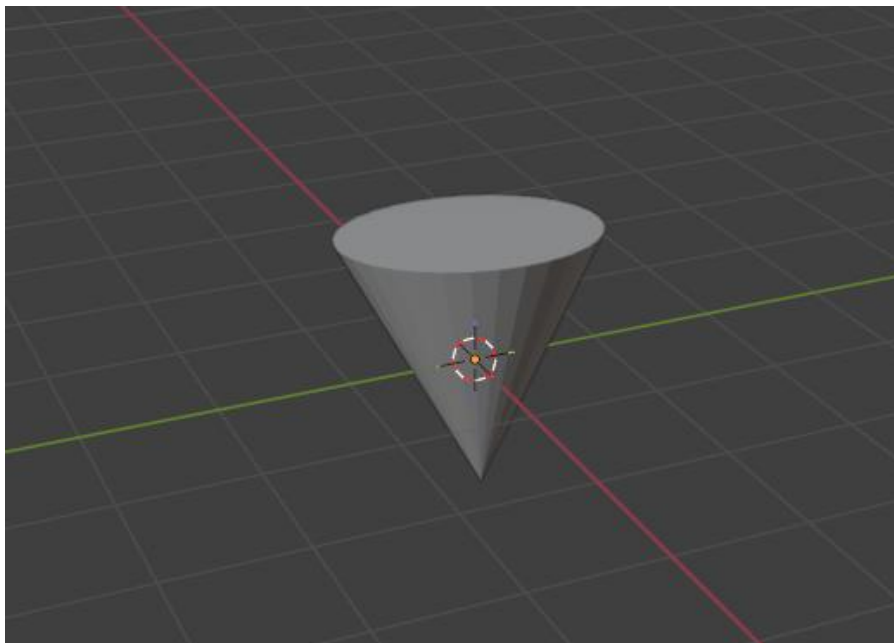
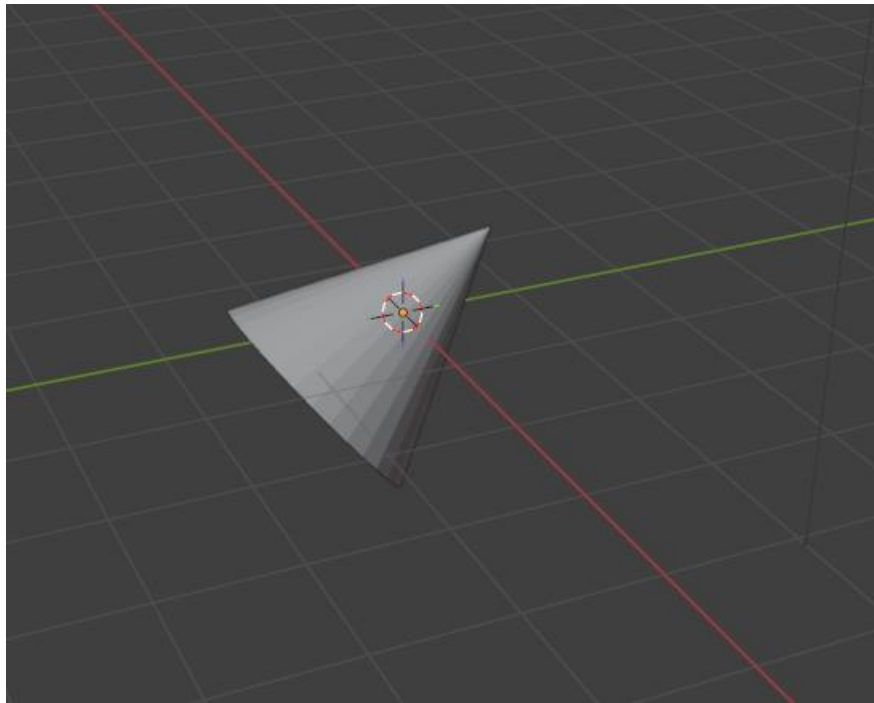
Scale the cube using SX, SY, SZ

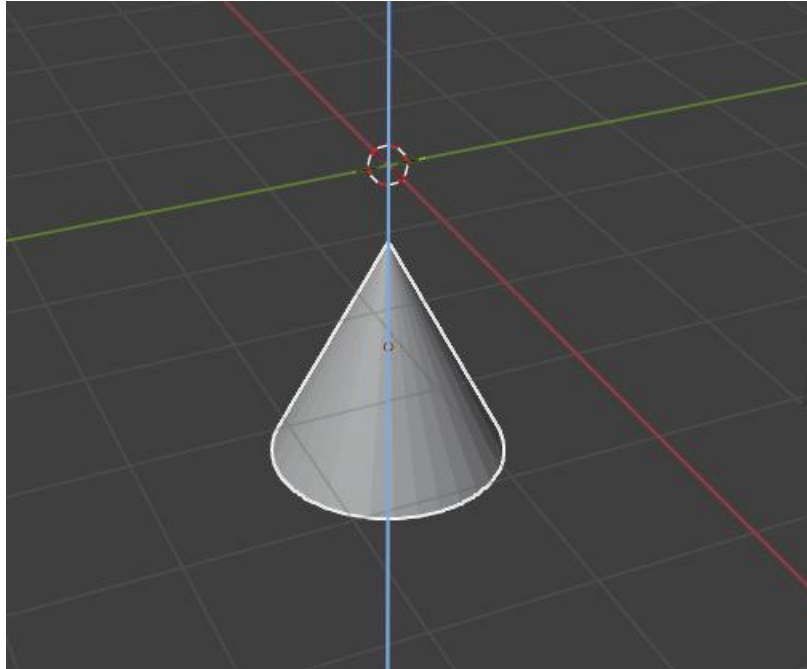




Task 3:

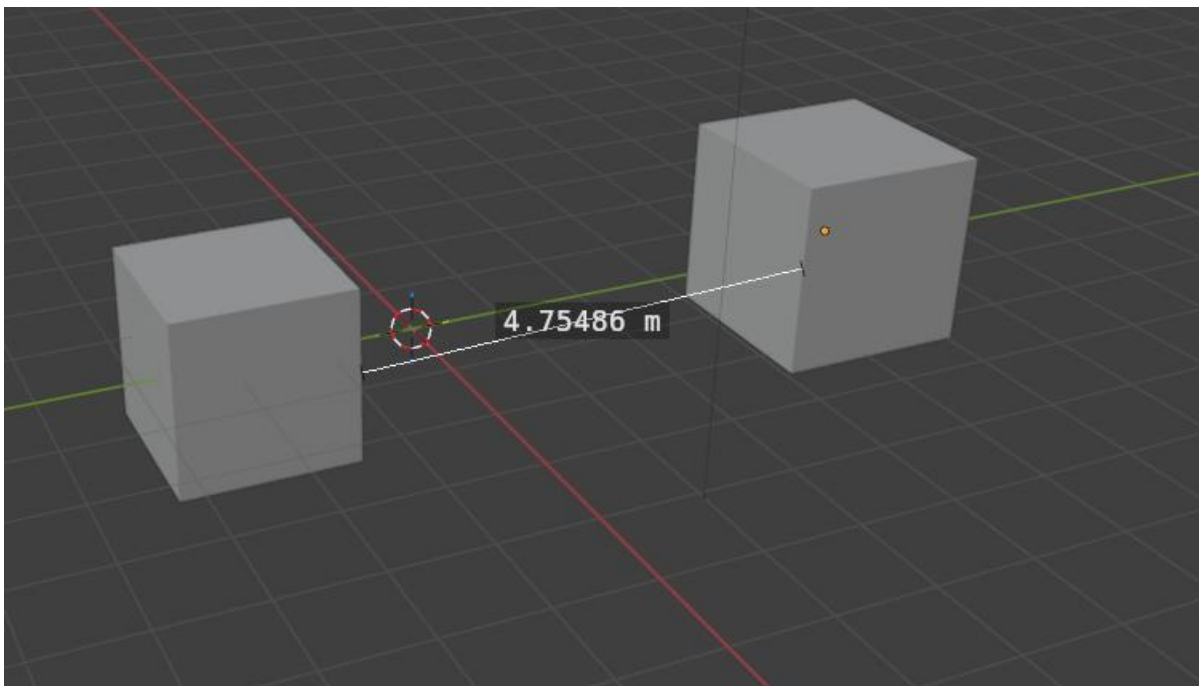
Rotate the cone on all axes (RX, RY, RZ)





Task 4:

Add another mesh object and measure distance



LAB 09

Blender Edit Mode (Basic Tools)

Objective

Students will learn:

- Edit Mode
- Edit Mode tools
- Vertex, Edge, Face selection

What is Edit Mode:

Edit Mode is used to:

- Edit the shape of an object
- Modify mesh structure

Shortcut:

- **Tab** → Switch between Object Mode & Edit Mode

Selection Modes:

- Vertex Select → **1**
- Edge Select → **2**
- Face Select → **3**

Basic Edit Mode Tools

- **Extrude** → **E**
- **Inset** → **I**
- **Loop Cut** → **Ctrl + R**
- **Delete** → **X**

Task 1

- Enter Edit Mode and select vertices (Vertex, Edge & Face)

Task 2

- Use Extrude (E) to extend object.

Task 3

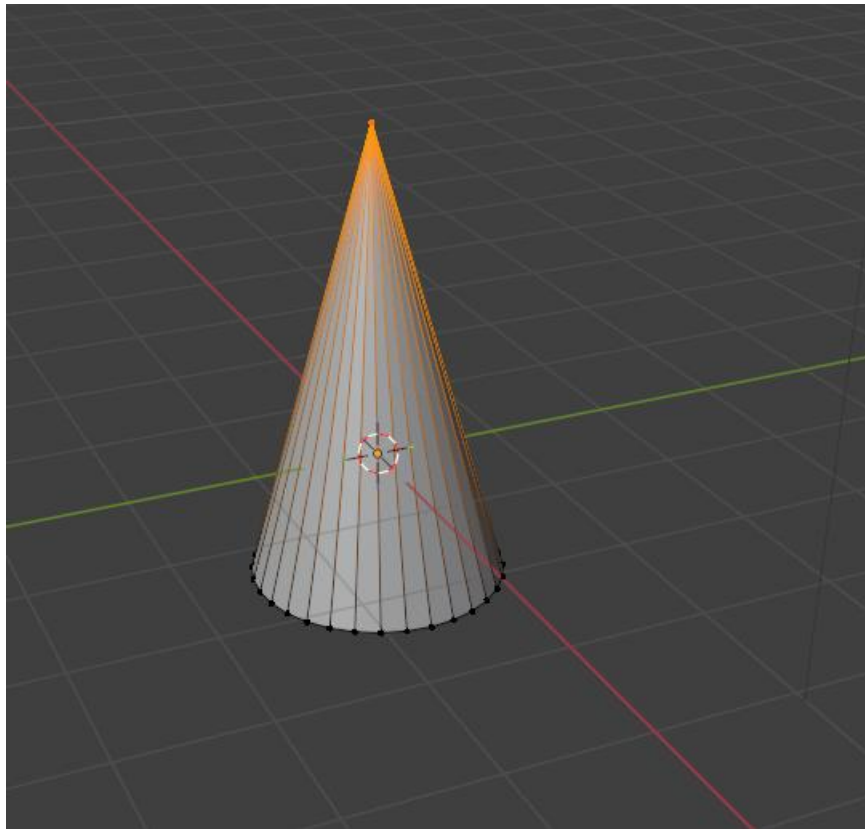
- Apply Loop Cut and modify shape.

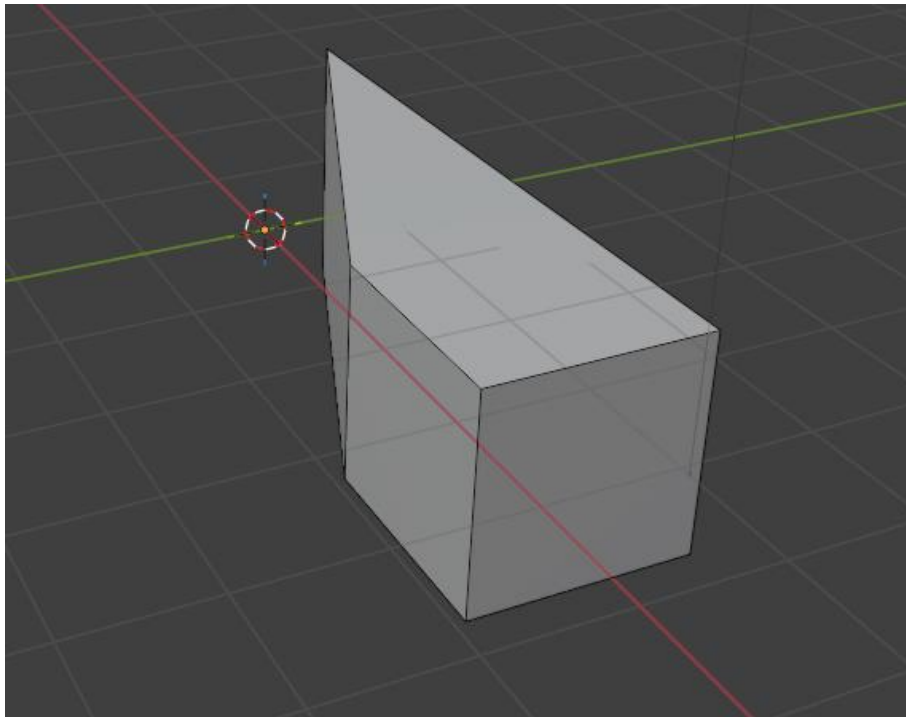
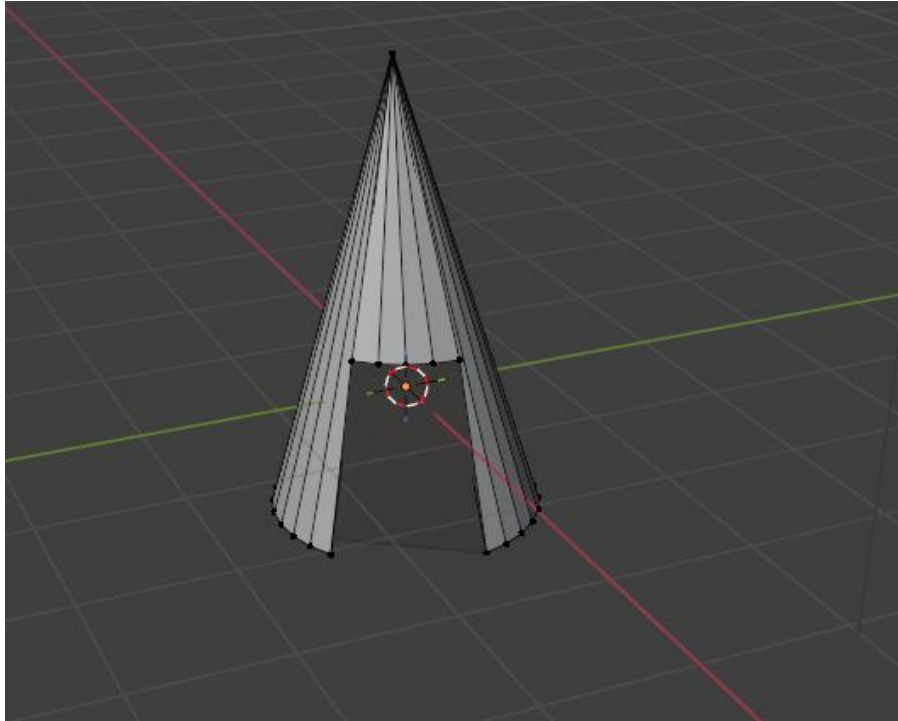
Task 4

- Apply Insert on shape.

Take Screen shots of all steps.

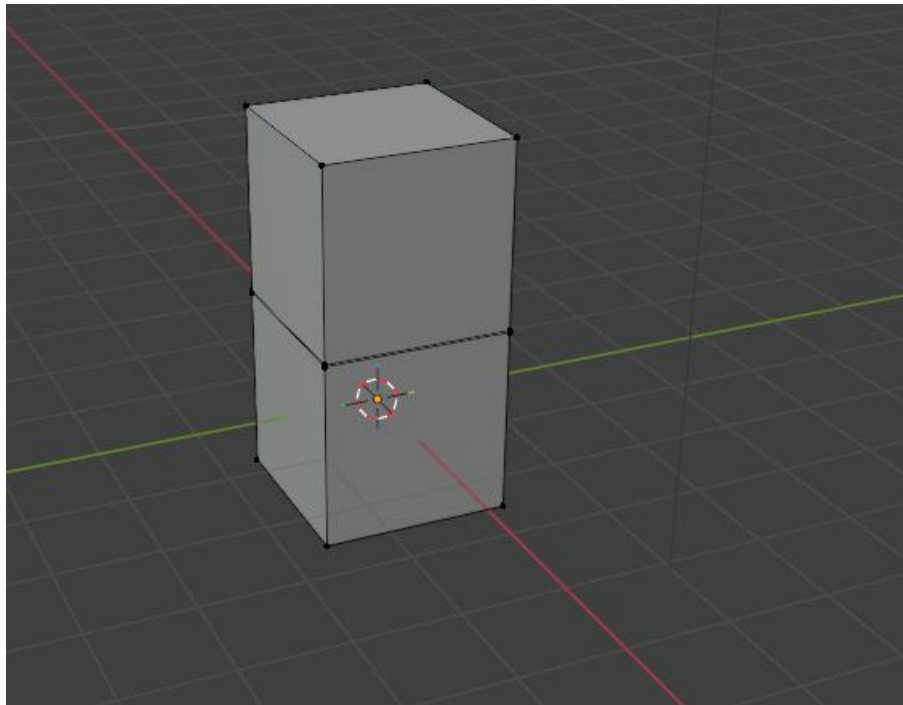
Task 1: Vertex, Edges and Faces





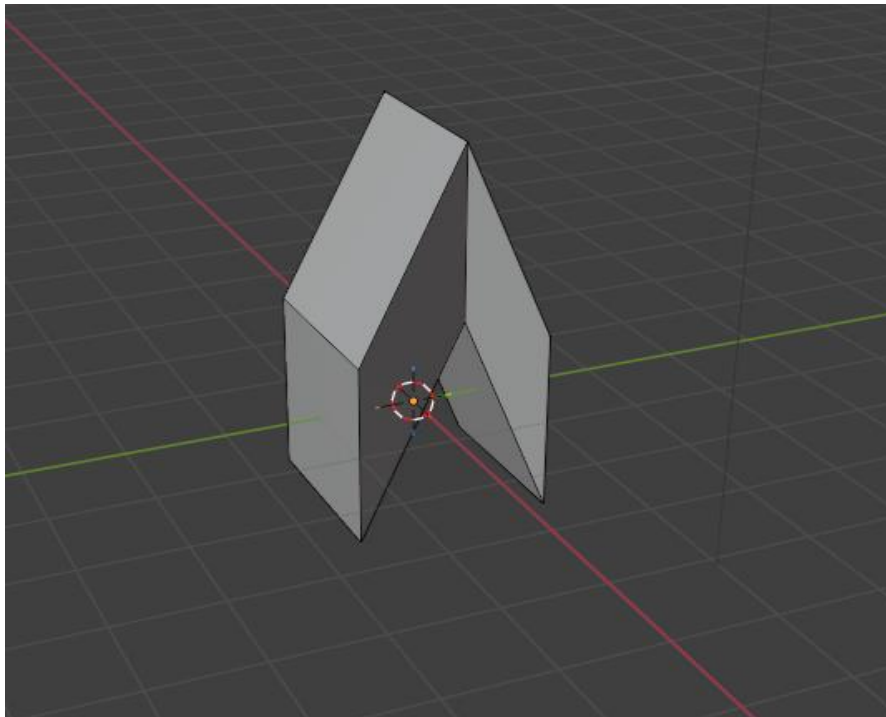
Task 2:

Use Extrude (E) to extend the object.



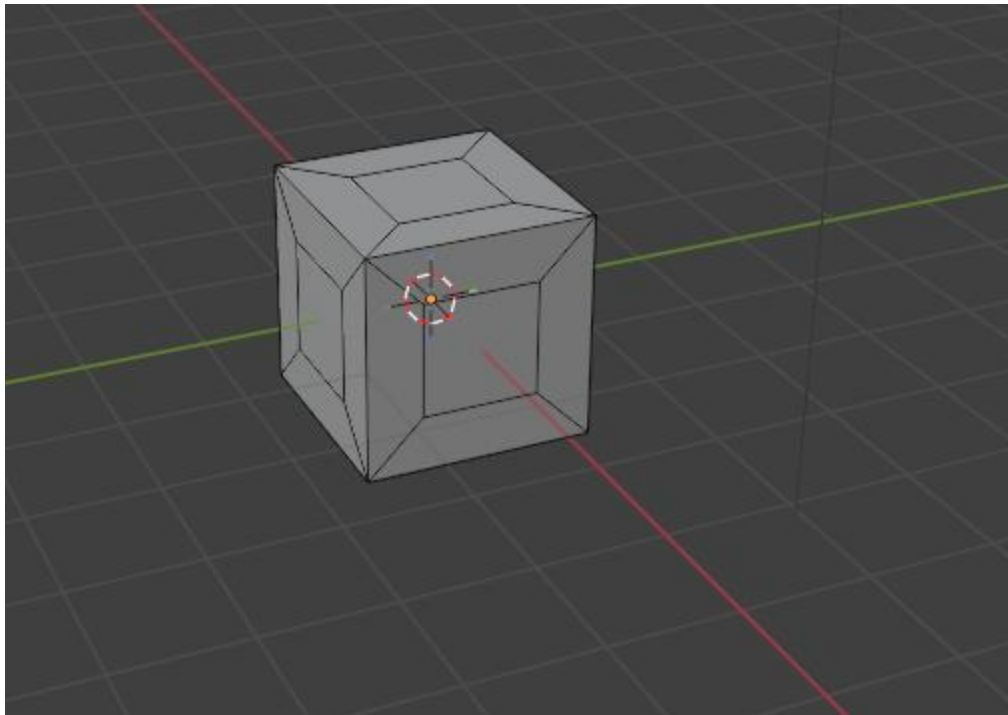
Task 3:

Apply Loop Cut and modify shape.



Task 4:

Apply Insert on shape.



LAB 10

Blender Edit Mode – Advanced Mesh Editing

Objective

By the end of this lab, students will be able to:

- Understand the purpose of **Edit Mode** in Blender
- Learn advanced mesh editing tools
- Modify vertices, edges, and faces professionally
- Create custom 3D shapes using mesh manipulation
- Improve object detailing using Bevel, Extrude, and Knife tools
- Practice non-destructive and accurate modeling techniques

Introduction

Edit Mode in Blender allows users to **modify the structure of a 3D object**. Unlike Object Mode (which works on the object as a whole), Edit Mode works on:

- Vertices
- Edges
- Faces

Advanced mesh tools help in:

- Making complex shapes
- Adding details to models
- Improving realism and structure

Switching to Edit Mode

1. Select any object (Cube recommended)
2. Press **Tab**
3. Edit Mode activates
4. Press **Tab** again to return to Object Mode

Mesh Selection Modes (Recap)

Mode	Shortcut	Use
Vertex Select	1	Edit points
Edge Select	2	Edit edges
Face Select	3	Edit surfaces

Advanced Edit Mode Tools

1. Extrude Tool

Shortcut: E

Purpose:

Used to **extend or pull out** vertices, edges, or faces to create depth.

Use Case:

- Creating walls
- Extending shapes
- Modeling doors/windows

Steps:

1. Select face / edge / vertex
2. Press **E**
3. Move mouse or use axis (X, Y, Z)

2. Bevel Tool

Shortcut: Ctrl + B

Purpose:

Smooths sharp edges by adding extra geometry.

Use Case:

- Making realistic objects
- Removing sharp corners

Steps:

1. Select edges
2. Press **Ctrl + B**
3. Scroll mouse wheel to increase segments
4. Click to apply

3. Loop Cut Tool

Shortcut: Ctrl + R

Purpose:

Adds **extra edge loops** for better control over shape.

Use Case:

- Dividing object
- Creating symmetry
- Adding detail

Steps:

1. Press **Ctrl + R**
2. Hover over object
3. Scroll to add loops

4. Click to confirm

4. Knife Tool

Shortcut: K

Purpose:

Allows **manual cutting** of mesh.

Use Case:

- Creating custom shapes
- Cutting irregular designs

Steps:

1. Press **K**
2. Click to start cutting
3. Click to end
4. Press **Enter** to apply

5. Merge Vertices

Shortcut: M

Purpose:

Combines two or more vertices into one.

Use Case:

- Cleaning mesh
- Reducing extra points

Options:

- At Center
- At Last

- At Cursor

6. Delete Geometry

Shortcut: X

Purpose:

Deletes selected mesh components.

Delete Options:

- Vertices
- Edges
- Faces

7. Inset Faces

Shortcut: I

Purpose:

Creates a smaller face inside an existing face.

Use Case:

- Window panels
- Door shapes
- Buttons

8. Fill Tool

Shortcut: F

Purpose:

Creates a face between selected edges or vertices.

Tasks:

Task 1: Shape Modification

- Add a Any Mesh
- Enter Edit Mode
- Use **Loop Cut** to divide it
- Apply **Bevel** on edges

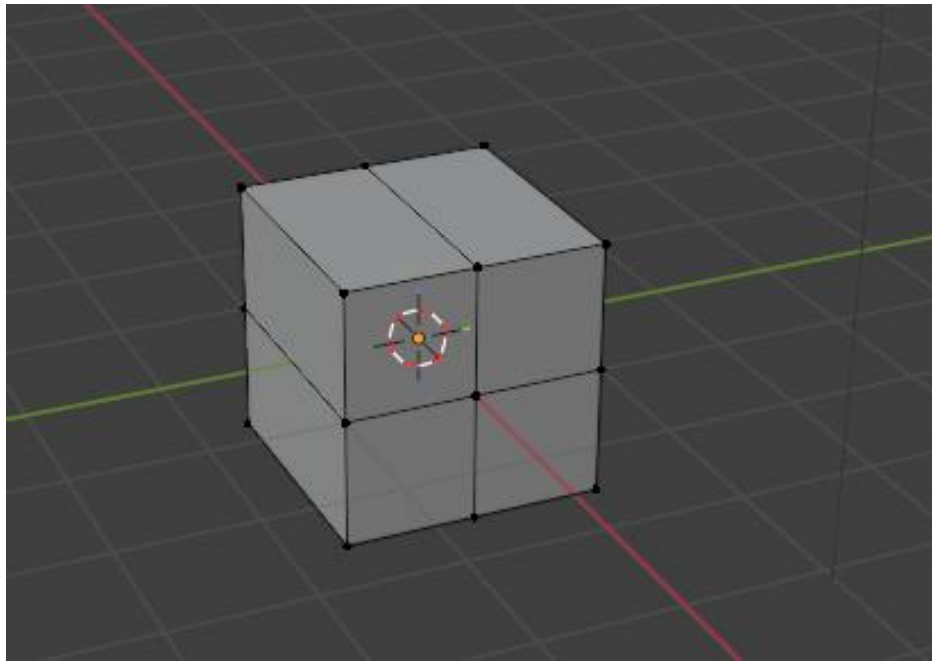
Task 2: Custom Design

- Add any Mesh
- Select a face
- Use **Inset (I)**
- Extrude (E) inward or outward

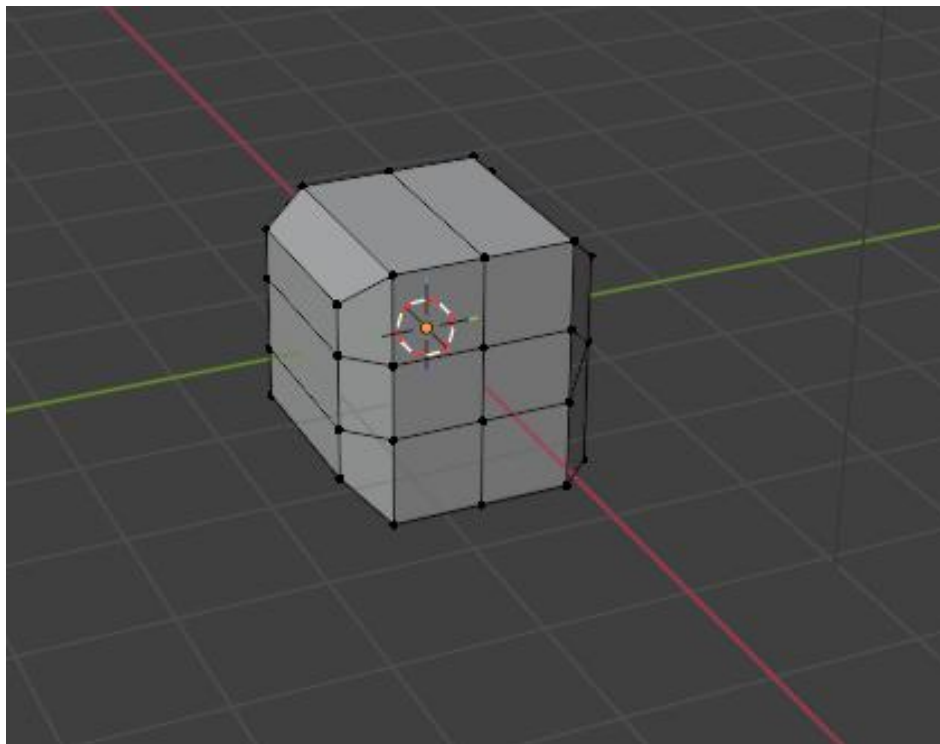
Take Screen shots of all Tasks.

Task 1: Shape Modification

Loop cut applied in the center both sides.

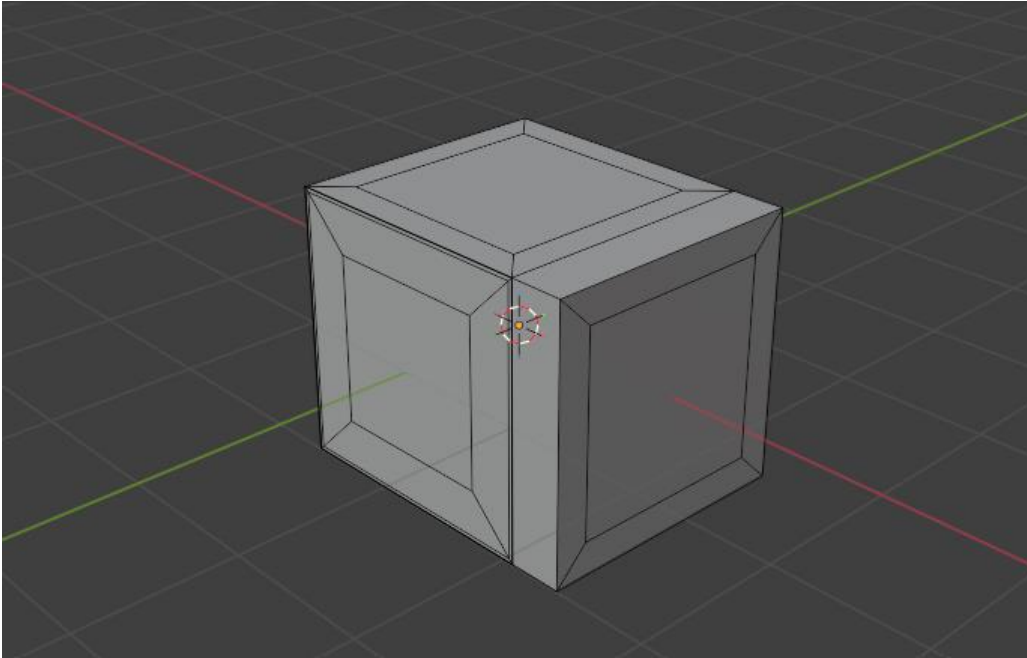


After applying Bevel on the loop cut mesh cube.



Task 2: Custom Design

Add any Mesh and select a face, and use **Inset (I)**



Extrude (E) inward or outward

